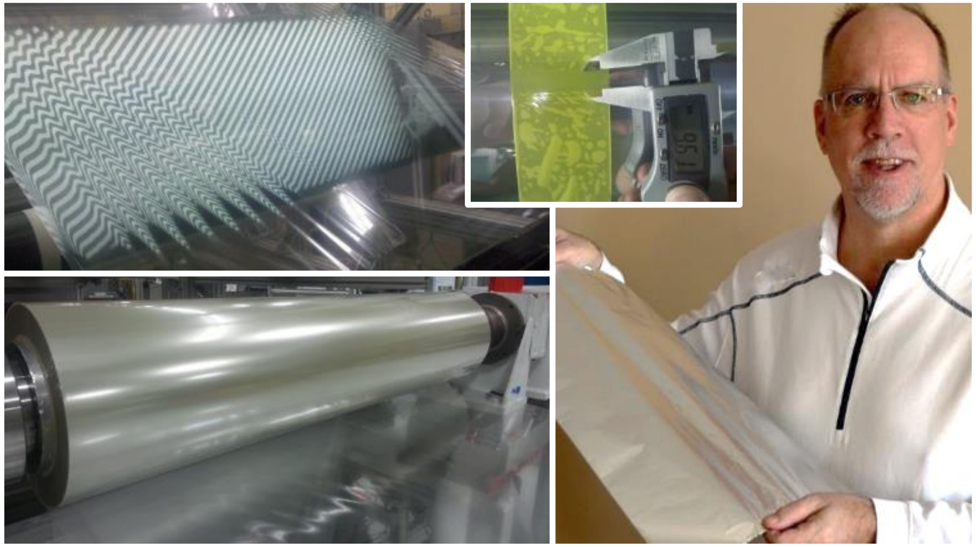
TJWALKER+ ASSOCIATES INC



1.0 What is Web Handling

Introduction and Overview

Timothy J. Walker, **TJWalker + Associates Inc.**

1620 Edgcumbe Road, Saint Paul, MN 55116

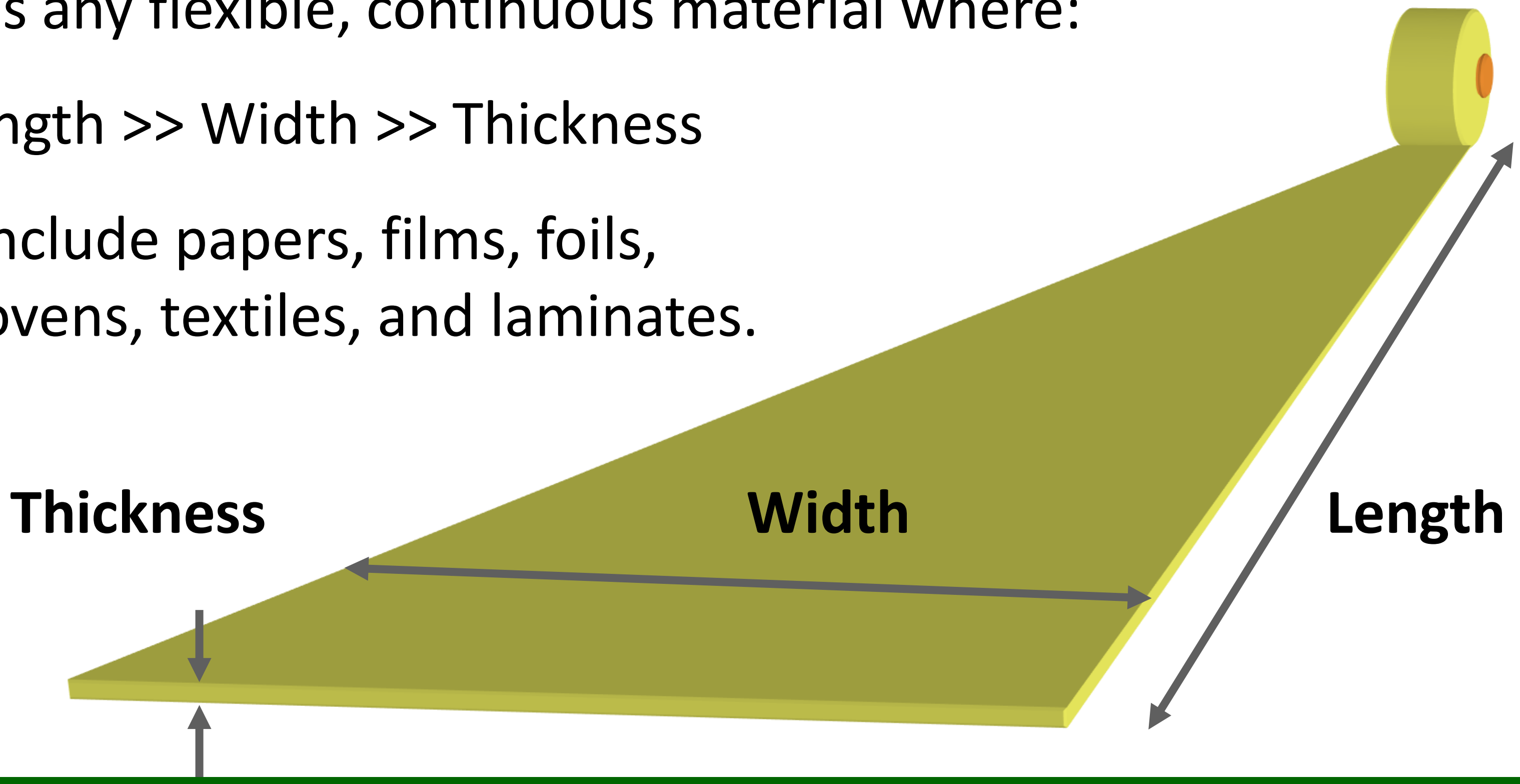
tjwalker@tjwa.com www.webhandling.com 651.686.5400

What is Web Handling?

- The process technology of transporting and storing thin, flexible materials.
- Flexible electronics called it roll-to-roll (R2R) processing.
- A web is any flexible, continuous material where:

  Length >> Width >> Thickness

- Webs include papers, films, foils, non-wovens, textiles, and laminates.



Web Handling Processes

Big: Paper Maker | Film Maker | Steel Pickling Line

Courtesy of Navigator Paper

Courtesy of Bruckner

Courtesy of Andritz

Small: Data Cartridge | Tape Applicator | Tape Applicator | Doctor Machine

Courtesy of 3M

Courtesy of ABI Systems

Courtesy of Accraply

Web Handling Processes

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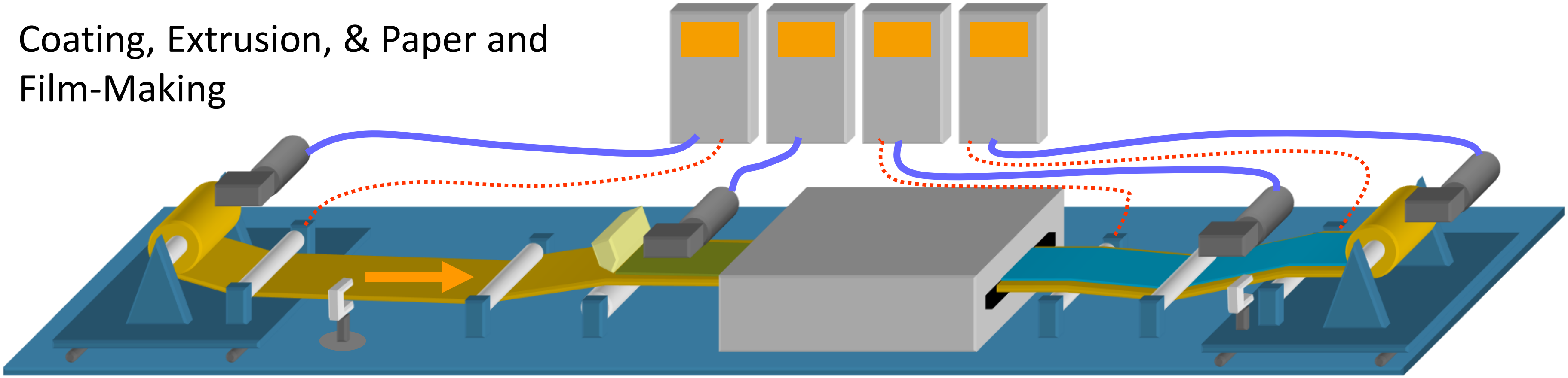
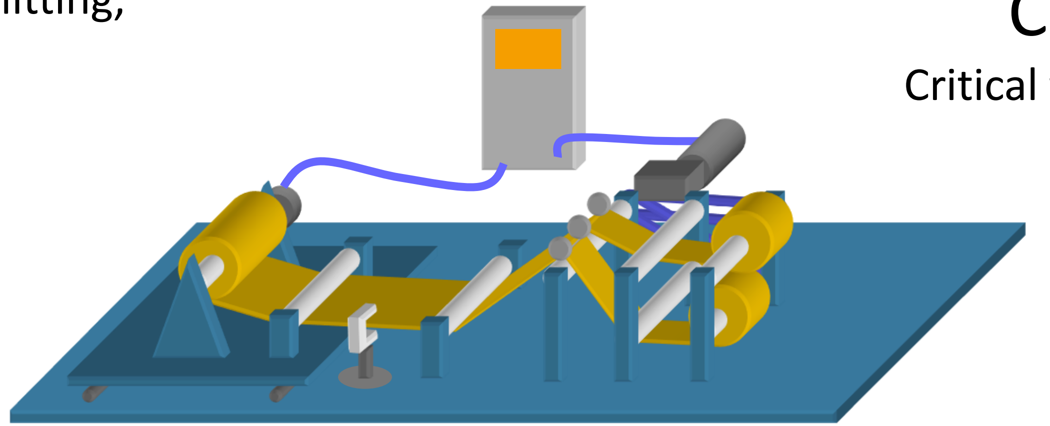
Makers:

Coating, Extrusion, & Paper and Film-Making



Converters: Slitting, Die-Cutting, & Laminating



Controls!

Critical to successful web handling.

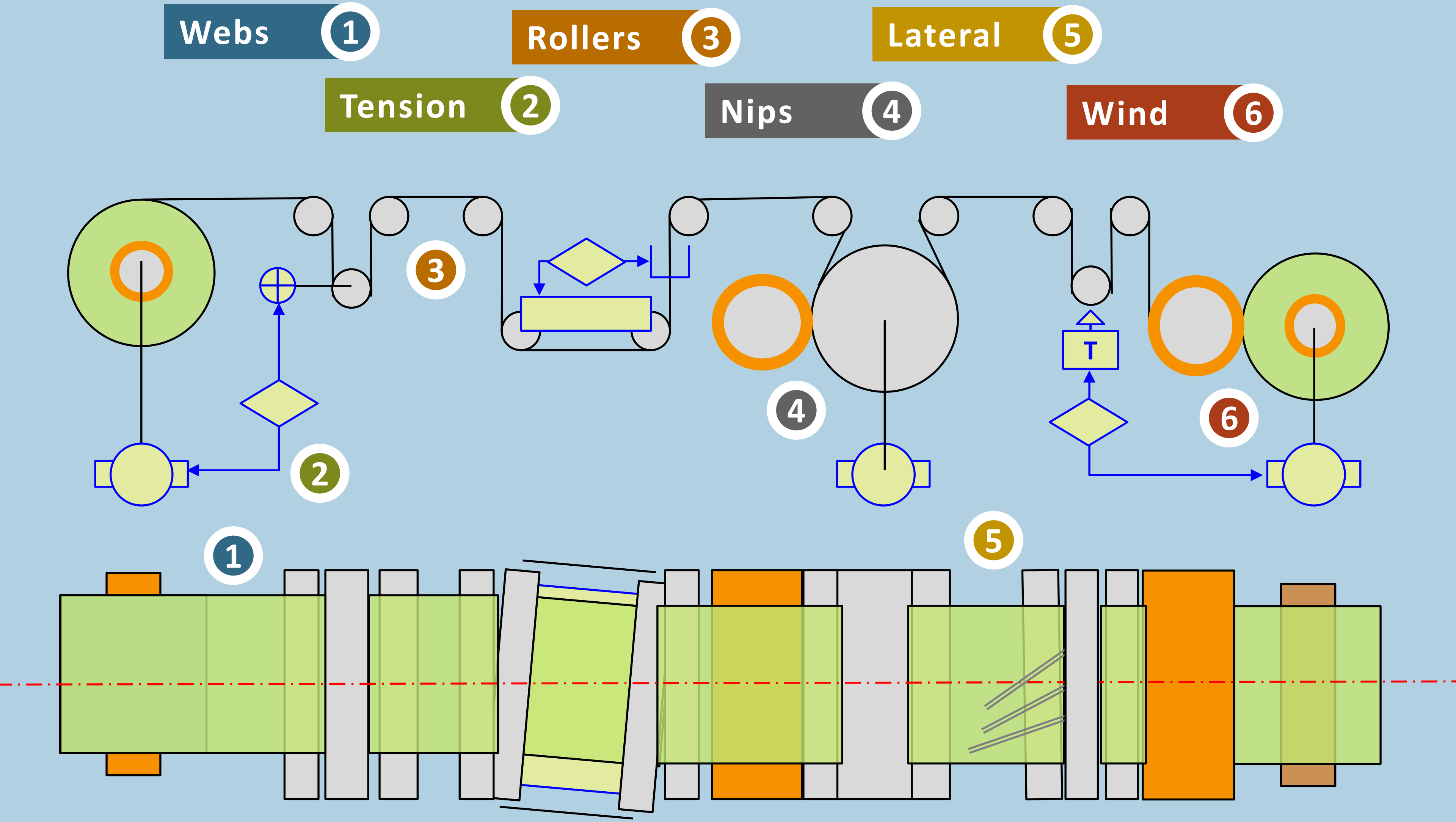
Web Handling Research Center – Oklahoma State University

Web Handling Research Center

- Mechanics of winding and unwinding
- Longitudinal dynamics and tension control
- Lateral dynamics and control; guiding and tracking
- Out-of-plane dynamics
- Wrinkling
- Air films between webs and rollers
- Aerodynamic effects in transport
- Measurement of tension, wound roll properties, and physical properties
- Precision lateral and longitudinal registration for roll-to-roll mfg.

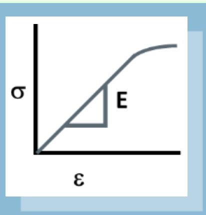
Founded in 1987. Held 13th Int'l Conf. on WH in 2015

Web Handling is...

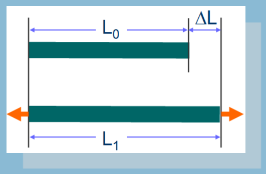


WH Fundamentals

1: Introduction to Webs and Web Handling



Web Properties of Tensioning

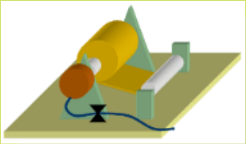


The Secret to Web Handling

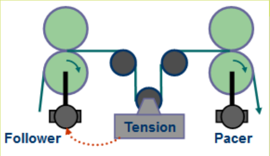
+ Web Properties
  - ✓ What material properties does a web handler need to know?
  - ✓ Know your web's mechanical and frictional properties.

WH Fundamentals

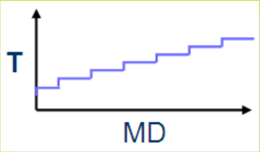
2: Tensioning



Tensioning Elements



Tension Control Systems

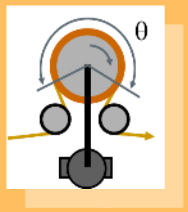


MD Tension Profile

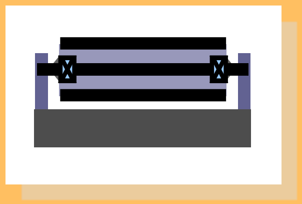
+ Tension Control
  - ✓ What is the right tension for your web or process?
  - ✓ What controls tension in your process?
  - ✓ What determines the optimum number of tension zones?

WH Fundamentals

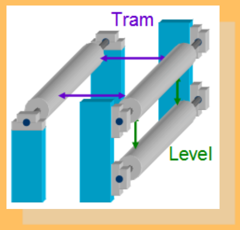
3: Rollers & Traction



Traction



Roller Basics



Alignment

+ Rollers and Traction

- ✓ Understand when webs slip on rollers
- ✓ Design rollers for driving, idling, and nipping applications
- ✓ Why is roller parallelism important?

WH Fundamentals

4: Nipping & Lamination

+ Nipping and Lamination
  - ✓ What is the load and pressure in a nip?
  - ✓ What causes or reduced pressure variations?
  - ✓ What are the best practices of lamination?

WH Fundamentals

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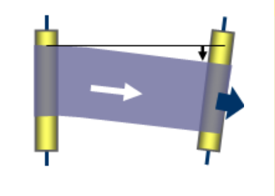
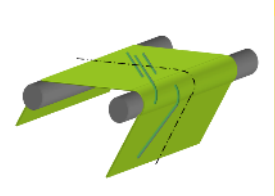
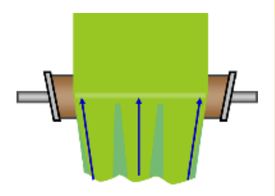
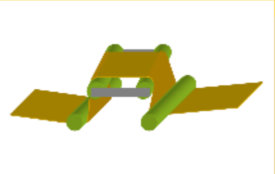
5: Lateral Motion: Tracking, Wrinkling, Spreading, Guiding


Tracking


Wrinkling


Spreading


Guiding

+ Lateral Motion
  - ✓ What cause a web to shift laterally?
  - ✓ What web guide is best for a process?
  - ✓ What causes web wrinkles?
  - ✓ What eliminates wrinkles?

WH Fundamentals

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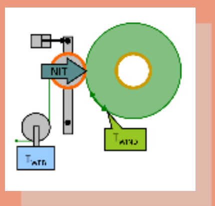
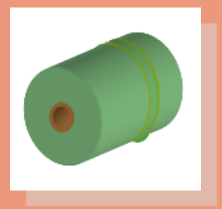
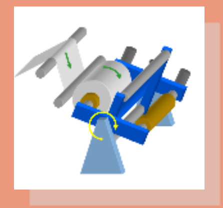
6: Winding: Process, Roll Quality, Equipment Design



Winding Process



Roll Quality



Winder Design

+ Winding, Rolls, Winders
  - ✓ What winder design is best for a product?
  - ✓ What determines the pressure inside a wound roll?
  - ✓ What causes roll and web defects?

WH Fundamentals

7: Web Handling Resources

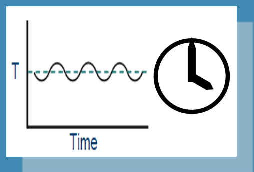
Resources

Web Handling On-Line Resources

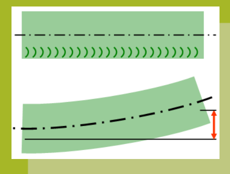
WH Fundamentals E-Book - Videos
TopWeb Software
PFFC Web Lines Column Archive - WH Resources

+ Resources
  - ✓ Where can I learn more about web handling?
  - ✓ Where can I get help with web handling?

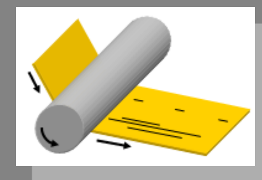
Web Handling and Winding Common Problem Areas



Tension Variations



Imperfect Webs



MD and CD Control

Web Buckling

1. Tensioning
2. Imperfect Webs
3. MD and CD Control
4. Web Buckling

Part 1: Solutions w/ Web Tensioning

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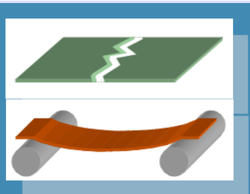
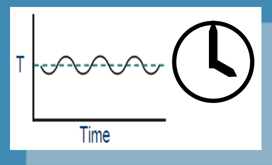
1A



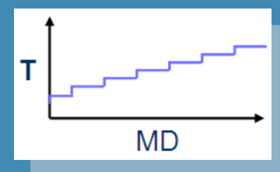
Tension: Too High or Too Low

1B



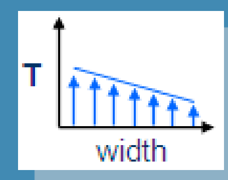
Tension: Varies Over Time

1C



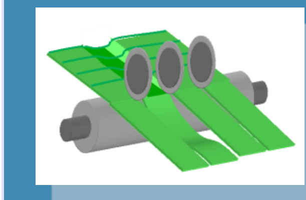
Tension: MD Variations

1D

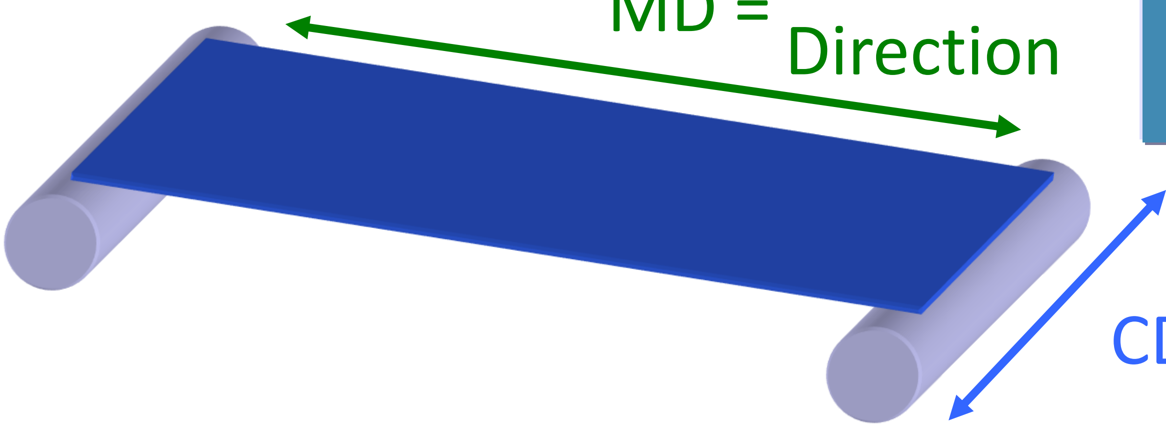


Tension: CD Variations

1E



Tension: Slit Strands

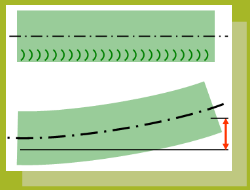


MD = Machine Direction

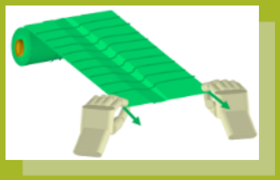
CD = Cross Machine Direction

Part 2: Solutions with Imperfect Webs

Cambered and Baggy Webs

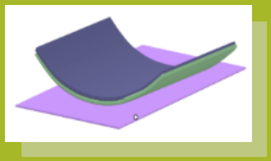


Cambered Webs



Baggy Webs

Non-Flat Webs

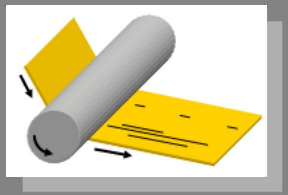


Curled Webs

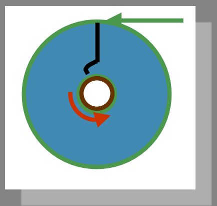
Part 3: Solutions with Control

Machine Direction Control

MD Slip Between Web and Driven or Idler Rollers



MD Slip on Rollers

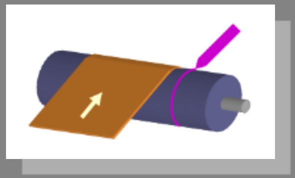


MD Slip in Rolls

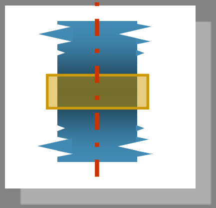
MD Slip Between Layers in a Rolls

Cross Machine Direction Control

Web CD Shifts or Slips Laterally on Rollers



CD Slip/Shift on Rollers

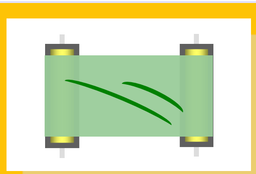


CD Error in Rolls

CD Shift or Misalignment of Layers in a Rolls

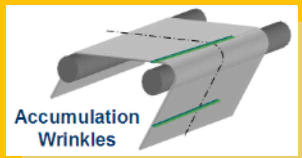
Part 4: Solutions with Buckled Webs

In Spans



Buckled Web in Spans

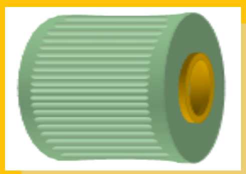
On Rollers



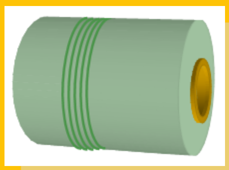
CD Buckles On Rollers

MD Buckles On Rollers

In Rolls



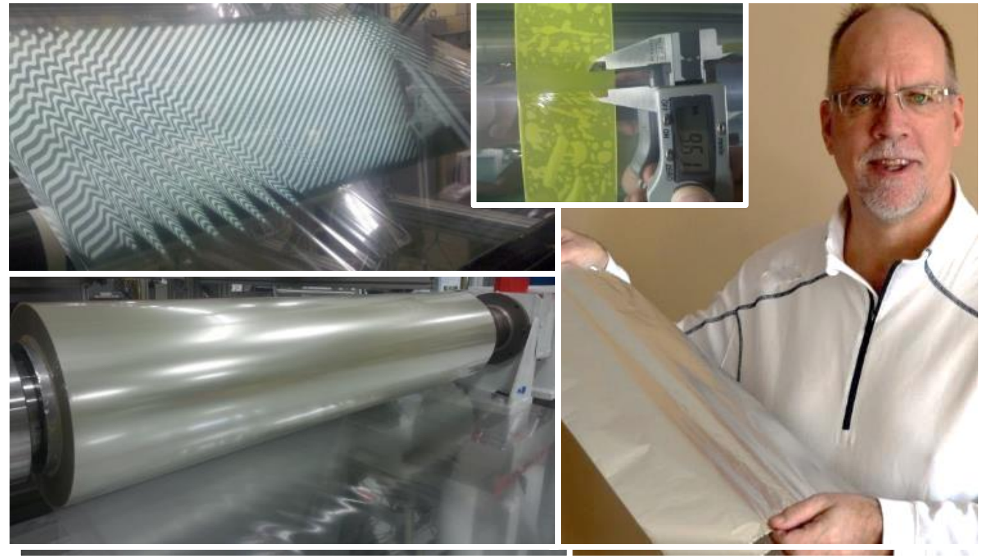
CD Buckles in Rolls



MD Buckles Rolls

Questions

- What do you want to learn?
- What questions do you have?
- What materials are you working with?
- What processes are you working with?
- What are your top sources of waste related to web handling?
- What would you like to share from your experience?



1.1 - Web Properties

What makes one web different from another?

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

tjwalker@tjwa.com www.webhandling.com 651.686.5400

What is Your Web? *Webatronium?*

- Thickness, Width
- Stress or Strain to Yield or Break
- Young's Modulus of Elasticity
- Surface Characteristics (Friction, Roughness, Porosity)
- Visco-Elasticity?
- Poisson's Ratio?
- Anisotropy? (MD-CD Differences)
- Layers? Laminate? Coated?

Films and Foils

- Polyethylene
  - Low-density (LDPE)
  - Medium-density,
  - High-density (HDPE)
  - Linear low-density (LLPE)
- Polypropylene
  - Cast film
  - Biaxially oriented film (BOPP)
  - Uniaxially oriented film.
- Polyester
  - BoPET (or PET) biaxially oriented polyethylene terephthalate film. (Mylar™, Melinex™, Estar™, Scotchpar™)
  - PEN (biaxially oriented polyethylene naphthelate)
- Polyimide
  - (Kapton™)
- BOPA
  - Biaxially oriented polyamide (Nylon™, Capran™)
- PVC (Polyvinyl chloride)
  - With or without a Plasticizer
- Cellulose triacetate
  - triacetate, CTA and TAC

- Aluminum
- Steel
- Copper
- Lithium

Many Types of Paper!


Bond Paper


Book Paper


Bristol Paper


Cover Paper


Index Paper


Offset Paper


Tag Paper


Text Paper

Courtesy of Okidata.com

For a long list of paper types:
http://www.paperonweb.com/grade.htm#a

Other “Webs”

- Non-wovens
  - Tyvek, fiberglass, spun bond, air laid,
- Textiles, Carpet
- Tissues
  - Bath tissue, facial tissue, toweling
- Scrim (open cloth)
- Belts (many materials)
- Meshes, Screens

What is a Non-Woven?

A fabric consisting of an assembly of textile fibres (oriented in one direction or in a random manner) held together (1) by mechanical interlocking; (2) by fusing of thermoplastic fibres, or (3) by bonding with a rubber, starch, glue, casein, latex, or a cellulose derivative or synthetic resin.

Mechanical Characterisation of Through-air Bonded Nonwoven Structures

Amit Rawal, Stepan Lomov, Thanh Ngo, Ignaas Verpoest and Jozef Vankerrebrouck

Thickness, Thickness Profile

How thick is a web?

- The answer to this is often a easy, but not always.
  - **Easy**: Foil, most films, coated papers
  - **Confusing**: Papers? Thickness can be measured, but are more often described by 'weight' (i.e. mass per area).
  - **Difficult**: Non-wovens, textiles, porous films (anything easily compressed).

Thickness, Thickness Profile

TJWALKER+ ASSOCIATES INC

What is a web's thickness profile?

- Thickness mapped vs. width.
- Does the web have thick edges? Lanes?
- Is the thickness profile consistent over time?
- Does the product have intentional thickness profile (including coatings or cutouts)?
- How is thickness profile measured? (Besides in winding defects)

Measuring Thickness

- Manual physical measurement

  Increase accuracy by measure a stack multiple sheets 10 layers, then divide by number in stack.

- Automated Off-Line
  - Beta radiation transmission
  - Capacitance
  - Physical contact (LVDT)
  - Mass per area weight sampling

Capacitance

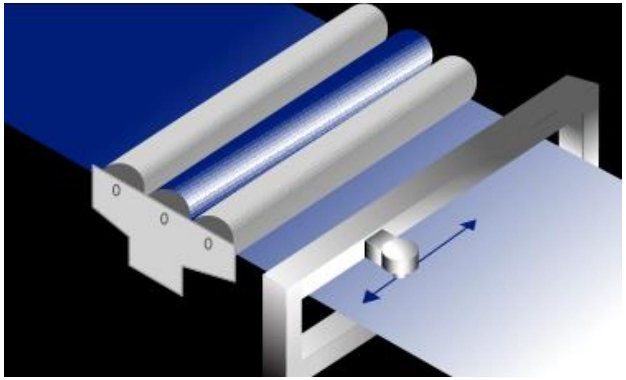
Combined

Absolute Contact

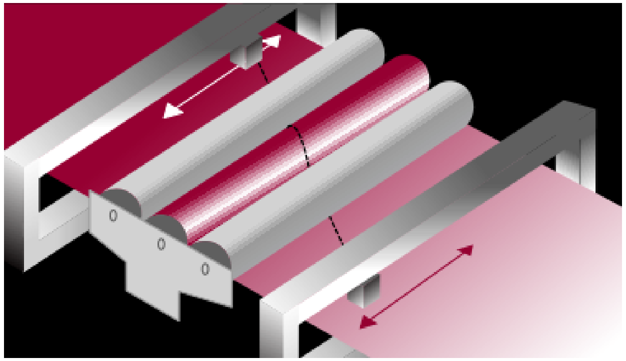
www.oaklandinstrument.com

Measuring Thickness

- Automated On-Line
  - Radiation (beta, x-ray) transmission
  - Radiation (gamma) backscatter
  - Near infrared (transmission, reflection)
  - Capacitance
  - Laser micrometer
  - Laser curtain on roller



Scanning



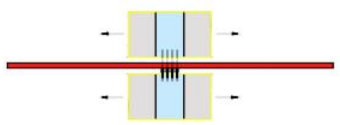
Same Spot Differential

www.ndc.com

Thickness Profile Measurement Options

X-RAY

SOFT X-RAY MEASUREMENT

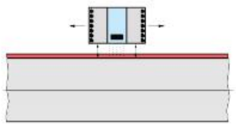


Non contact systems with O frame

max. thickness depending on material
No legal permission required.
No special radiation safety officer required

KAPA

CAPACITIVE / EDDY CURRENT

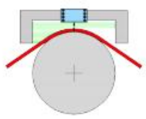


Non contact systems with measurement roll

max. thickness 2 mm

SHADOW

LASER SHADOWING

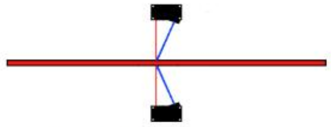


Non contact systems with measurement roll

max. thickness 5 mm depends on sheet stiffness

STG

LASER DISPLACEMENT



Non contact systems with O frame

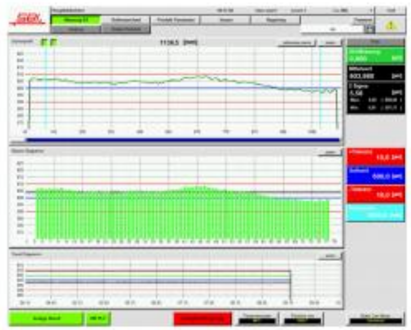
sheet thickness up to 40 mm

www.sbi.at

Closed-Loop Thickness Profile Control

Thickness profile measurement is often combined with software and controls to make automatic adjustments of extrusion or coating dies.

SOFTWARE

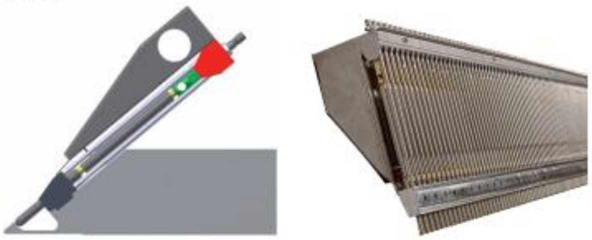


CONTROLS

- Thickness control
- Thickness profile control
- Lip gap adjustment
- Fine-tuning of film / sheet properties

THERMAL BOLTS

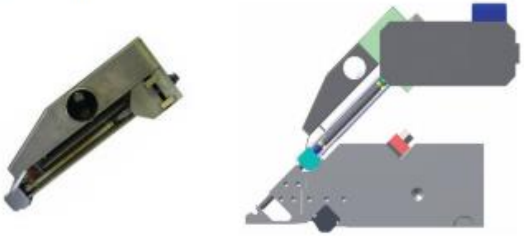
Heated thermal bolts for thickness profile control in film /sheet extrusion.



SBI Automatic system for extrusion dies

FRAD - Range: 2 mm -

Heated bolts combined with mechanical adjustment system (electromotive driven).

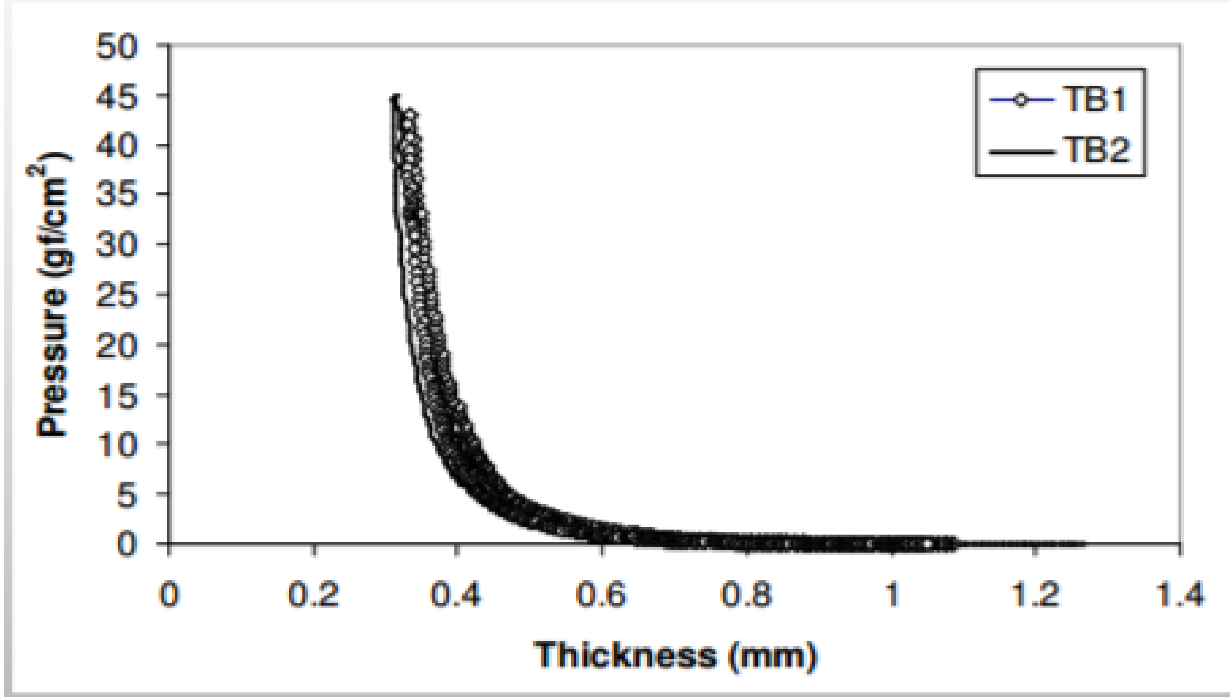


2 mm adjustment range for SBI automatic extrusion-die system FRAD

www.sbi.at

What is the Thickness of a Non-Woven?

The thickness of a nonwoven or tissue is difficult to quantify, as even low pressures exerted on a sample will compress it to a reduce thickness.



Mechanical Characterisation of Through-air Bonded Nonwoven Structures

Amit Rawal, Stepan Lomov, Thanh Ngo, Ignaas Verpoest and Jozef Vankerrebrouck

Paper Weights vs Thickness

"Basis Weight" is defined as the weight of 500 sheets of paper in its basic unit uncut size before being cut to Letter size or Legal size. The most common sizes are Bond, Text, Book, Cover, Index and Tag.

An uncut sheet of Bond paper is 17 x 22 inches and Cover paper is 20 x 26 inches.

- If 500 sheets of Bond paper (17 x 22 inches) weigh 20 lbs, then a ream of paper cut to Letter size will be labeled as 20 lb.
- If 500 sheets of Cover paper (20 x 26 inches) weigh 65 lbs, then a ream of this paper trimmed to tabloid size is marked as 65 lb.

Courtesy of Okidata.com

Basis Weight vs. Caliper Thickness (Estimated)

US Basis Weights				Caliper	Metric
Bond	Text	Cover	Index	1 Point = 0.001"	GSM (g/m^2)
20	50				75
24	60				90
28	70				105
32	**80**				120
36	90	**50**			136
38	**100**	55		6.0	140
43	110	**60**	**90**		162
47	**120**	**65**	97	8.0	177
54		74	**110**		199
58		**80**	120	10.0	218
		90	135		245
		93	**140**	12.0	253
		100	150		271
		115	**170**	14.0	310
		130	200	16.0	350

Courtesy of Okidata.com

Paper Basis Weight, Thickness, Density, Bulk

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- Density = mass / volume — g/cc
- Bulk = volume / mass (inverse of density) — cc/g
- Basis weight = mass / area — g/cm^2
  - (area = length x width)
- Density = basis weight / thickness — g/cc
- Bulk = thickness / basis weight — cc/g

- Water: 1 g/cc density, 1cc/g bulk

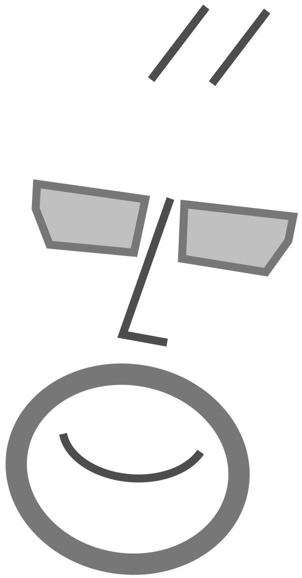
- cc = cubic centimeter = cm^3 = $0.000001m^3$

The Right Tension Qs

What is the right tension?

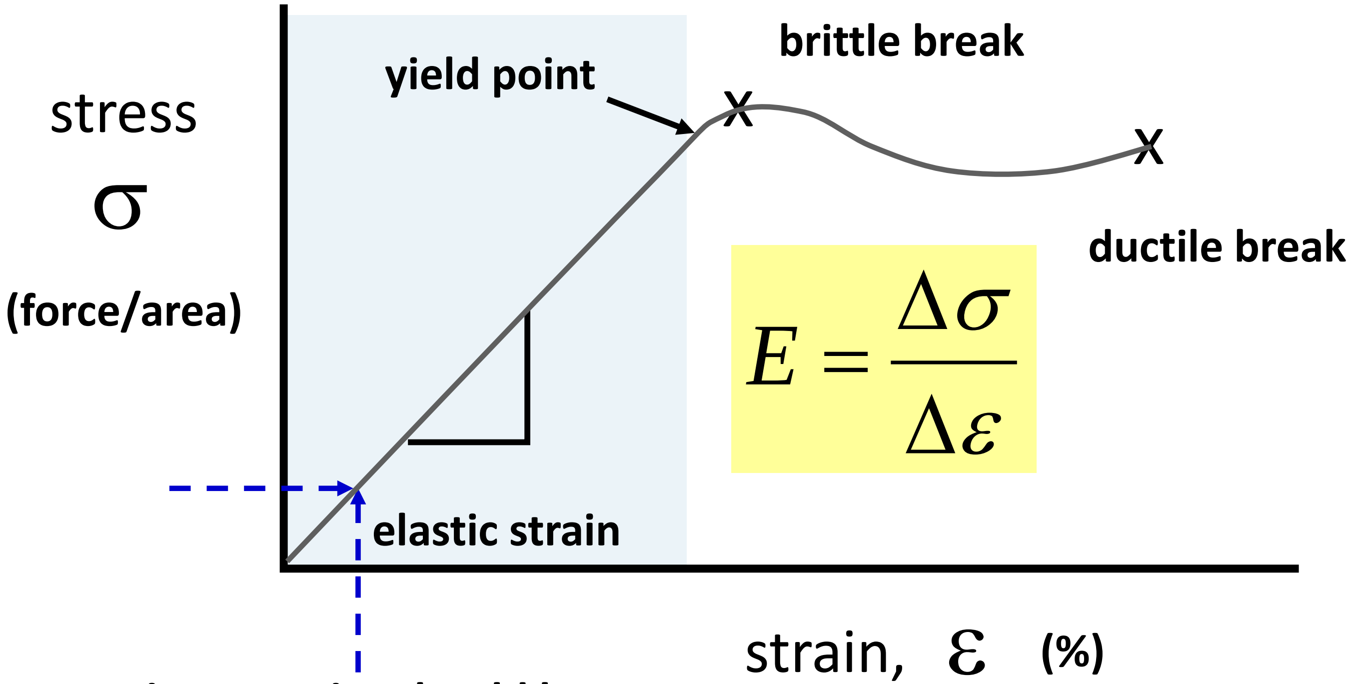
The answer to this requires understanding of:

force, tension,
stress, strain,
tensile-elongation testing,
yield point, break point,
& modulus of elasticity.



Web Handling Stress

Typical web handling tensions are 10 to 20% of a web's yield or break stress.



Tension setpoint should be 10-20% of yield or break point.

Common Units & Variables

t	= web thickness, mils or inches,	0.001” = 25μm
w	= web width, inches,	1” = 25.4mm
L	= length, inches,	1” = 25.4mm
V	= web or roller speed, feet/min. (fpm),	100 fpm = 30 m/min
D	= diameter	
r	= radius	
F_T	= Force of tension, lbs,	1 lbs = 0.454 kg
T	= tension force per width, pli, (pli = Pounds / Lineal Inch of width)	1 pli = 175 N/m
σ	= web stress, psi,	1 psi = 6.9 kPa
ε	= web strain, dimensionless	
E	= Young’s modulus, psi,	100,000 psi = 690 Mpa
ν	= Poisson’s ratio	dimensionless
μ	= coefficient of friction or traction	dimensionless
ρ	= density, lbs/in^3	1 lbs/in^3 = 0.277 g/mm^3
θ	= misalignment angle, degrees or radians	1 mil/in = 1mm/m = 1 mradians
β	= wrap angle, degrees or radians	180 degrees = 3.1415 radians

Tension Units (SI)

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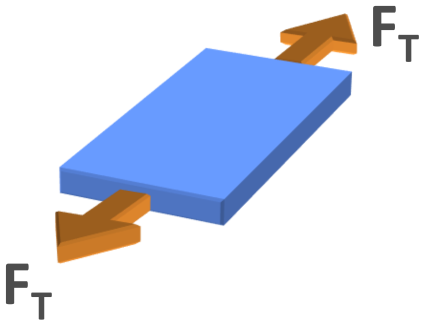
F_T

Tension Force

Force

lbs (kgf or N)



F_T = 10 kgf

F_T ~ 100 N

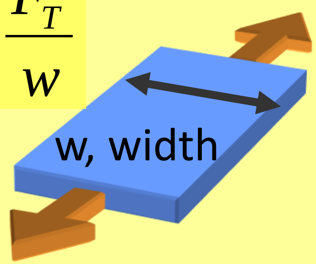
T

Tension

Force per Width

pli (kgf/m)

$$T = \frac{F_T}{w}$$



F_T = 10 kgf

w = 0.5 m

T = 20 kgf/m

T ~ 200 N/m

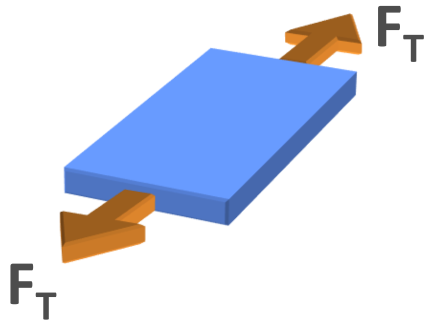
Tension Units (English)

F_T

Tension Force

Force

lbs



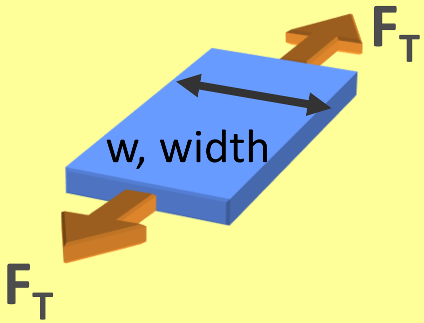
F_T = 10 lbs

T

Tension

Force per Width

pli



F_T = 10 lbs

w = 10 in.

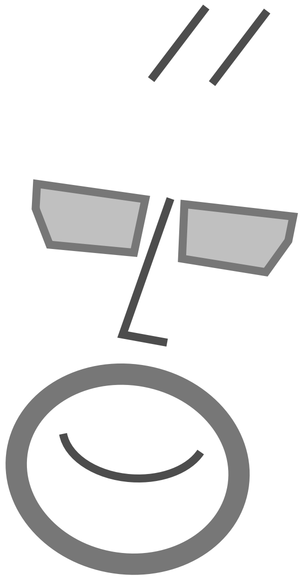
T = 1 lbs/in

T = 1 pli

The Right Tension Qs

What tension will break or deform a web?

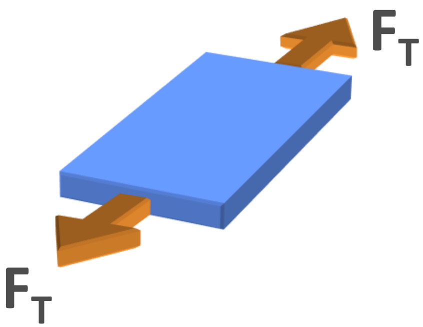
What is stress?
What is strain?
What is modulus?



Tension Units (SI)

TJWALKER+ ASSOCIATES INC

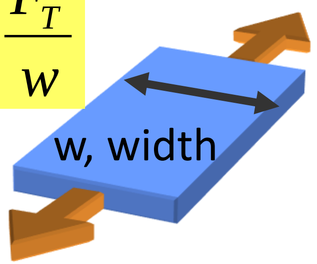
F_T
Tension Force
Force
lbs (kgf or N)



F_T = 10 kgf
F_T ~ 100 N

T
Tension
Force per Width
pli (kgf/m)

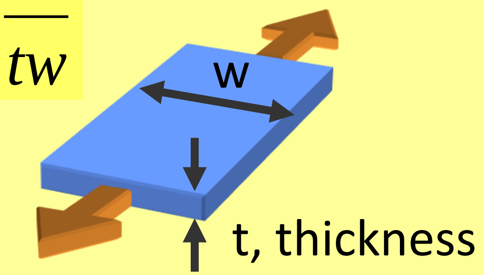
$$T = \frac{F_T}{w}$$



F_T = 10 kgf
w = 0.5 m
T = 20 kgf/m
T ~ 200 N/m

σ_T
Tensile Stress
Force per Area
lbs/in^2 (kPa)

$$\sigma = \frac{F}{tw}$$



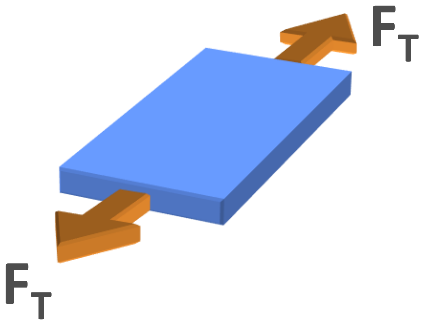
F_T = 100 N
w = 1 m
t = 25 μm
σ_T = 4 MPa

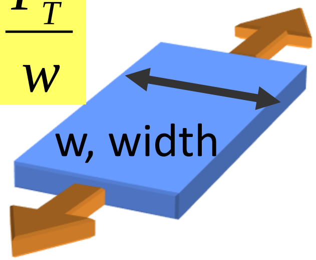
Tension Units (English)

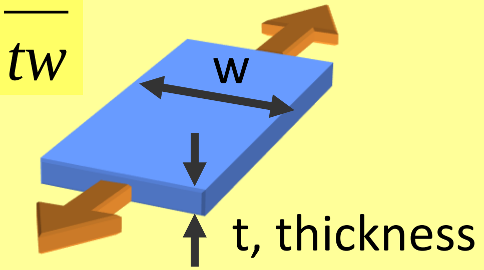
F_T	T	σ_T
Tension Force	Tension	Tensile Stress
Force	Force per Width	Force per Area
lbs	pli	lbs/in²

$$T = \frac{F_T}{w}$$

$$\sigma = \frac{F}{tw}$$







F_T = 10 lbs

F_T = 10 lbs
w = 10 in.
T = 1 lbs/in
T = 1 pli

F_T = 10 lbs
w = 10 in
t = 0.001 in
σ_T = 1000 psi

KEY CONCEPT

Stress

Stress and pressure are both defined in units of force per areas. To better understand any web process, convert forces into stresses by dividing the load by the cross-sectional area it is exerted over.

Use: Machine tension is commonly measured as a force in units of lbf, kgf, or N. To compare different products or processes, calculate tensile stress by dividing tension force by product thickness and width.

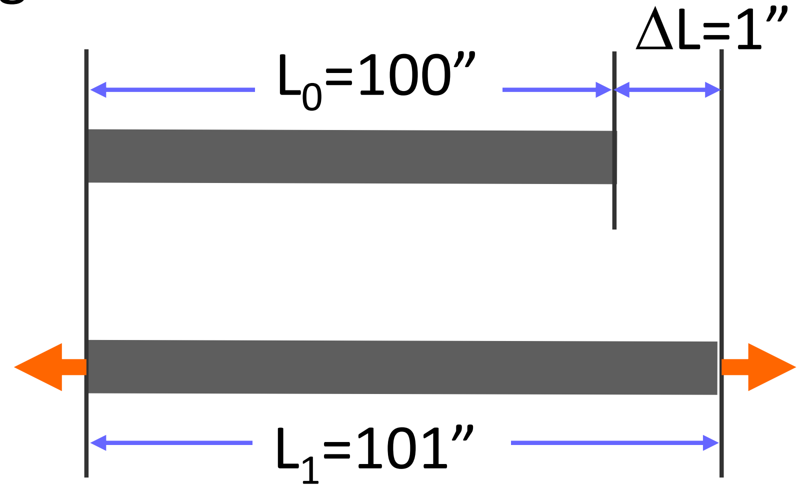
Example:
F_T = 50 lbf creates higher stress as cross-sectional area decreases.
For w=50” and t=0.010”, the tensile stress is a low 100 psi.
For w=50” and t=0.001”, the stress is 1000 psi.
For w=1” and t=0.001”, the stress is 50,000 psi!

Strain Defined

Strain, ε, is the ratio of the change in a dimension over the untensioned dimension. For tensioning, strain is the change in length over the untensioned length.

L_0 = Dimension of zero tension web

L_1 = Dimension of tensioned web



$$\varepsilon = \frac{L_1 - L_0}{L_0} = \frac{\Delta L}{L_0} = \frac{101-100}{100} = 0.01 = 1\%$$

KEY CONCEPT

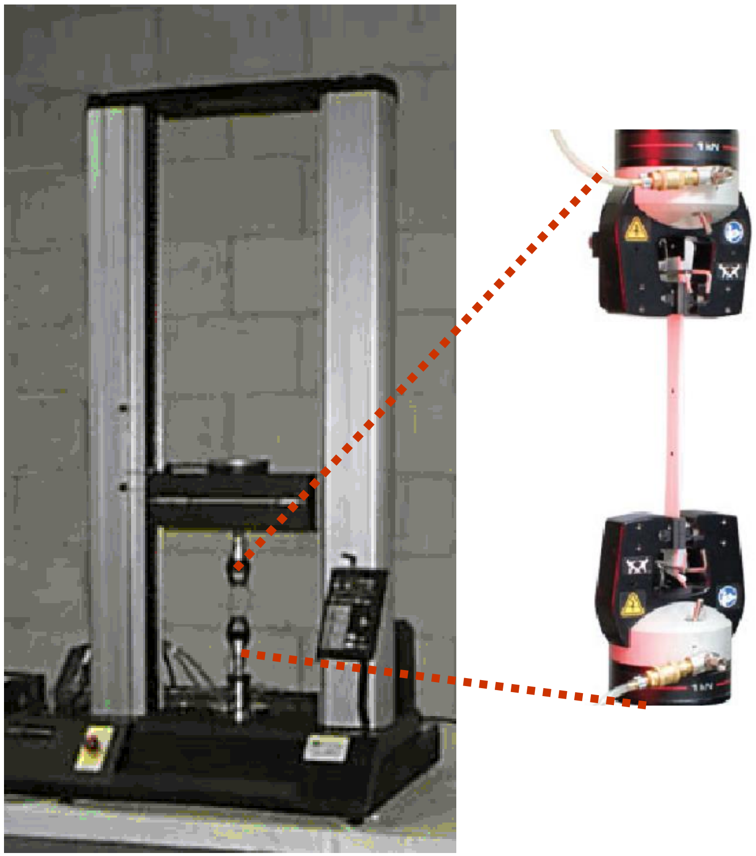
Strain

Strain is dimensional change in a solid material in reaction to stress. For positive stresses, materials will elongate in the direction of the stress. For negative stress or pressure, materials will compress. Strain is calculated as the change in dimension divided by the original dimension.

Use: When a web is forced to conform around variations in roller parallelism or diameter, the web's response will begin by determining the web strain.
Example: If a roller diameter varies from 5.00" to 5.05", the web develop a 1% strain differential to conform to roller.

Tensile-Elongation Testing

Most tensile-elongation testing focuses on measuring ultimate break strength or elongation, but should provide other mechanical properties.

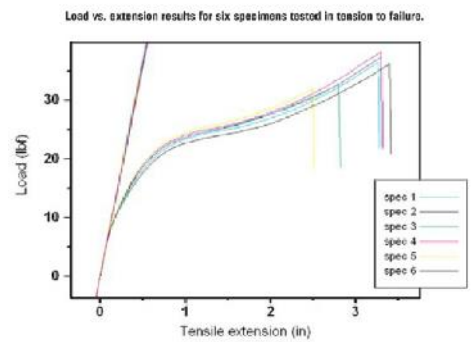


Specimen size:
ASTM D882 uses 1" x 6" strips die cut from thin sheet or film.

Data:
The following calculations can be made from tensile test results:

1. tensile strength (at yield and at break)
2. tensile modulus
3. strain
4. elongation and percent elongation at yield
5. elongation and percent elongation at break

Tension or Force



%Elongation

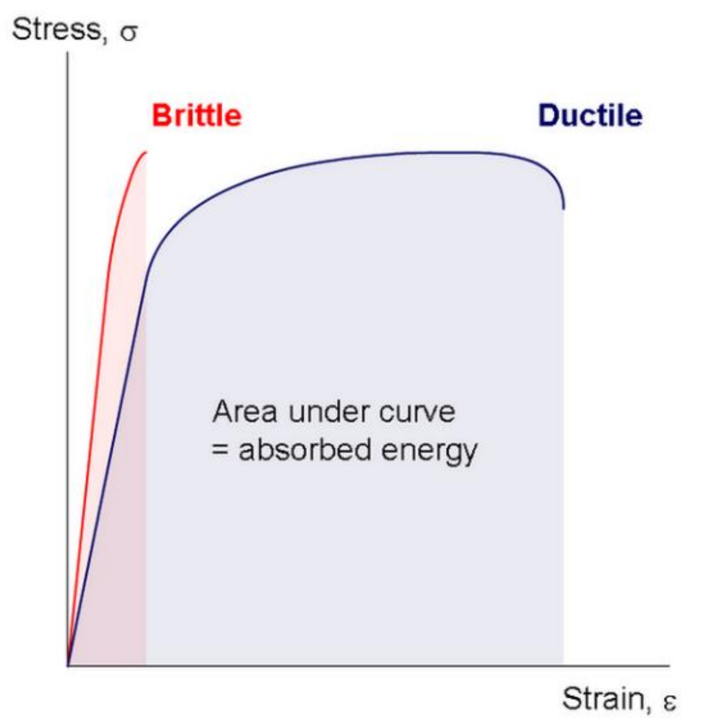
https://www.youtube.com/watch?v=7zlXSQ1OFf4

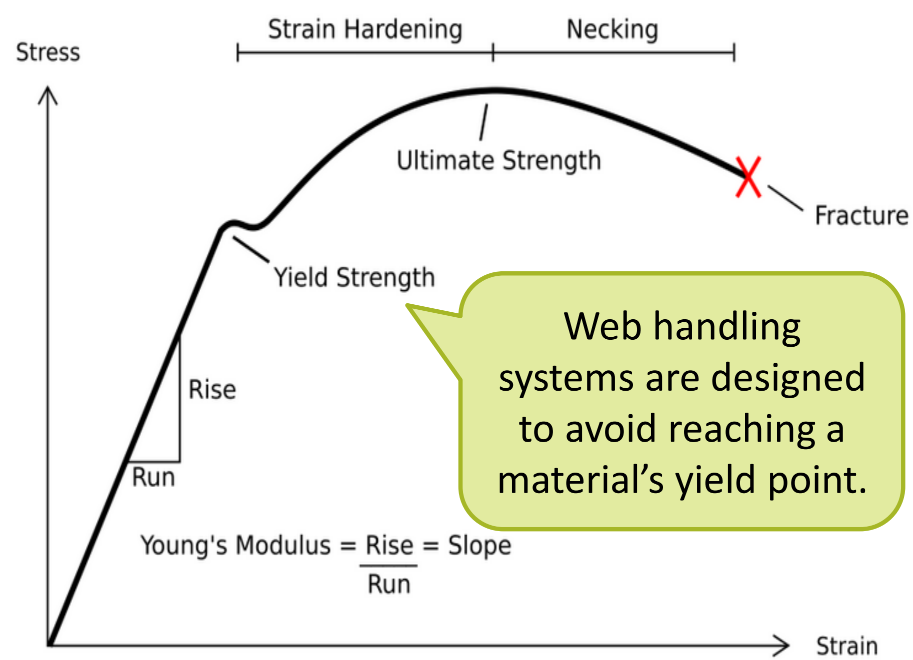
Images courtesy of Instron

Ductile vs Brittle

In brittle materials, fracture occurs without noticeable change in the elongation rate.

Ductile materials are able to yield at normal temperatures. The yield point where a material begins to deform plastically. Prior to the yield point the material will deform elastically and will return to its original shape when the applied stress is removed. Beyond the yield point, some portion of the deformation will be permanent and non-reversible.

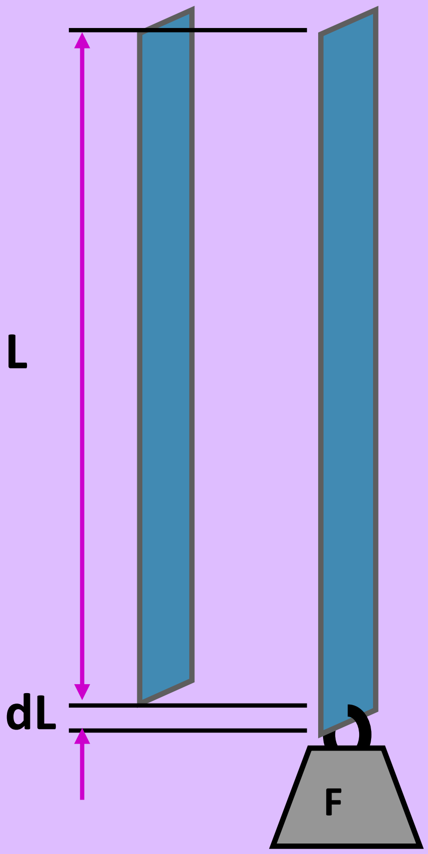




Web handling systems are designed to avoid reaching a material's yield point.

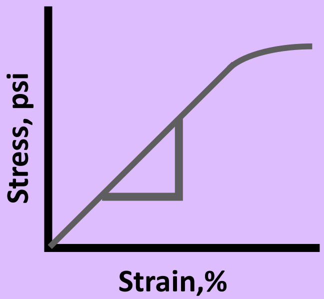
Measuring Modulus

Image courtesy of OSU WHRC



Calculate modulus by noting the length change of a strip under a known weight (tension).

$$\sigma = \frac{F}{tw}$$

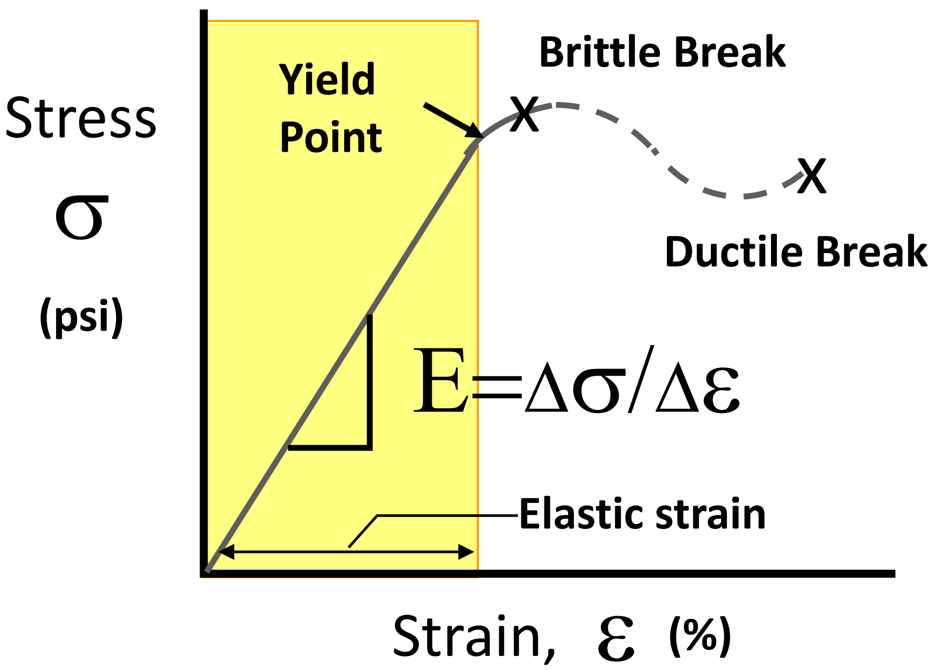


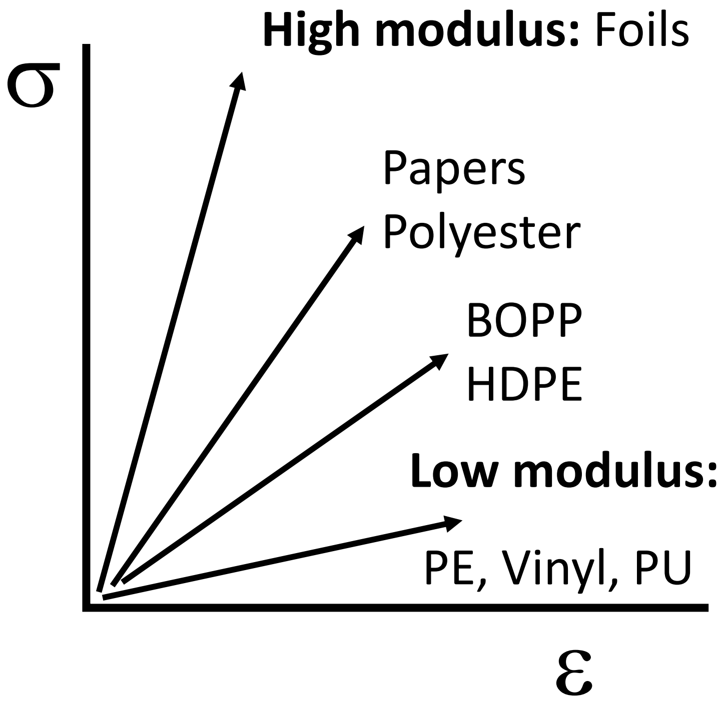
$$\varepsilon = \frac{\delta L}{L}$$

$$E = \frac{\delta\sigma}{\delta\varepsilon} = \frac{FL}{(\delta L)tw}$$

Modulus Defined

Modulus is the initial slope the stress-strain curve.
It describes a web's "stretch-ability."

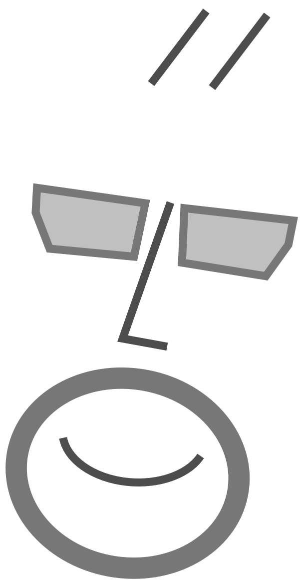




Modulus is Important

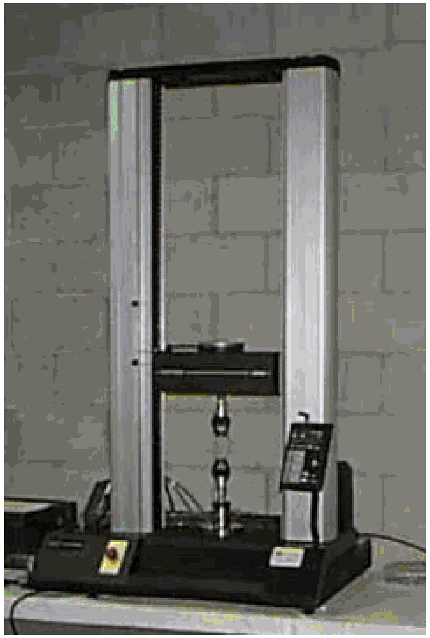
It is quite common to find quality labs measuring break or yield points, but never calculating modulus.

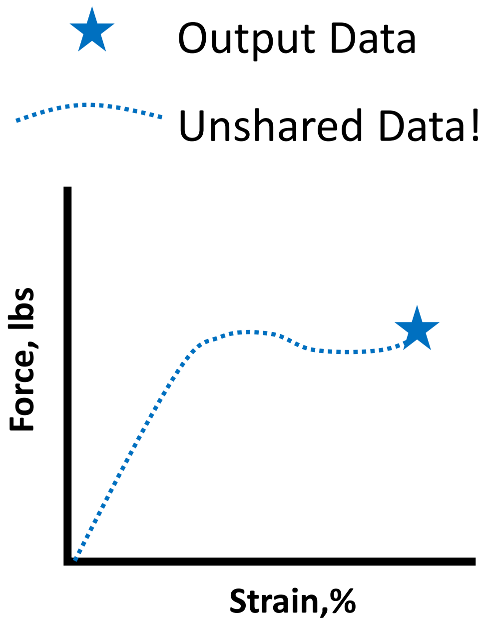
Knowing modulus is quite useful in web handling and critical to laminating.



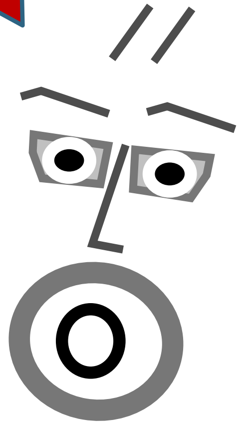
Modulus? We Don't Do Modulus!

Quality labs commonly have tensile elongation testers for break, peel , or tear testing, but when asked to measure modulus, 1) have never done it, 2) don't know how to, and 3) find no help in the equipment manual...



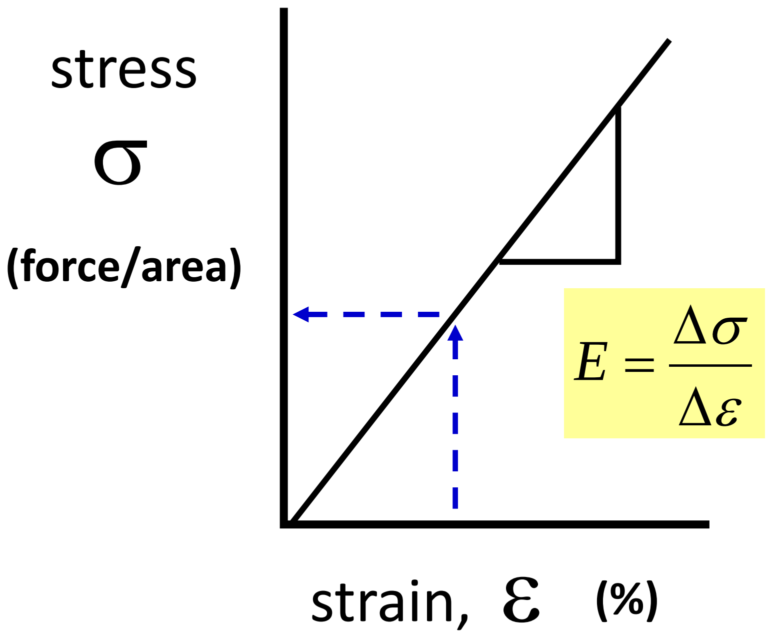


= very frustrated web handler.



Using Elastic Modulus

The elastic modulus is a material property that describes the relationship of stress to strain.



$$\sigma = \varepsilon E$$

ε = 0.01 = 1%
E = 500,000 psi
σ = εE = (0.01)(500000) = 5000 psi
= 5PLI / mil

$$\varepsilon = \frac{\sigma}{E}$$

σ = 0.5 PLI / 0.5 mil = 1000psi
E = 200,000 psi
ε = σ/E = (1000)/(200000) = 0.005
= 0.5%

Tension/Strain Ratio

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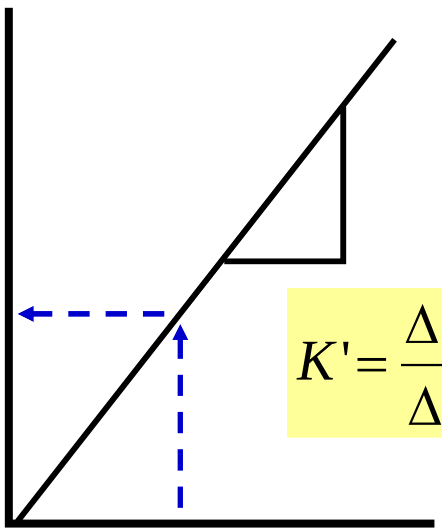
For webs with difficult to define thickness, tension to strain ratio can be a useful replacement for modulus.

Tension

T

(force/width)



$$K' = \frac{\Delta T}{\Delta \varepsilon}$$

strain, ε **(%)**

ε = 0.01 = 1%
K' = 50 PLI
T = εK' = (0.01)(50) = 0.5 PLI

$$T = \varepsilon K'$$

T = 2 PLI
K' = 40 PLI
ε = T/K' = (2)/(40) = 0.05 = 5%

$$\varepsilon = \frac{T}{K'}$$

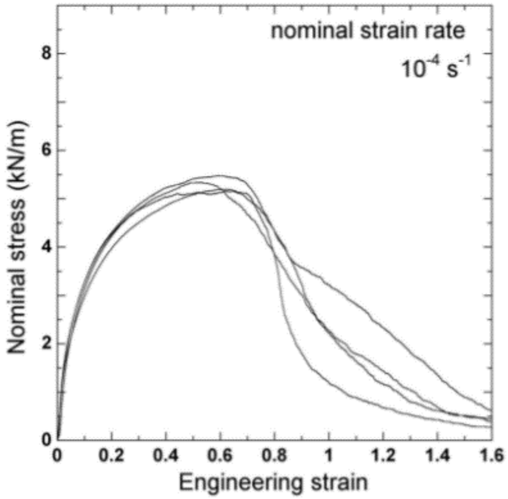
Example Web Mechanical Properties

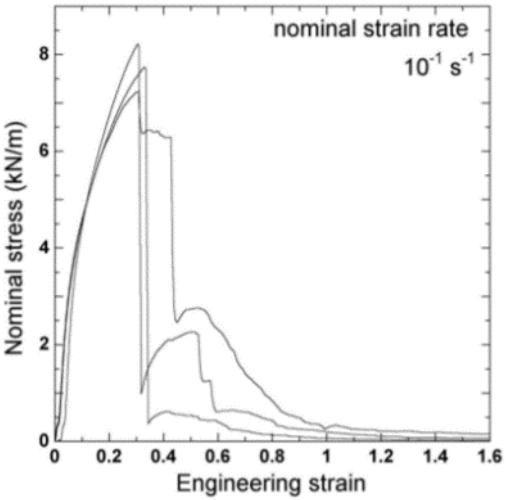
	E	ρ	σ_{yield}	T/t	ε_{yield}	E	ρ	σ_{yield}	T/t
Material	Modulus	Density	Yield Stress	Target Tension	Yield Strain	Modulus	Density	Yield Stress	Target Tension
	(GPa)	(kg/m^3)	(MPa)	(N/m/μm)	(%)	(Mpsi)	(lb/ft^3)	(kpsi)	(PLI/mil)
Steel	200	7800	300	30.0	0.15	29	486	43.5	4.4
Aluminium	69	2710	165	16.5	0.24	10	169	23.9	2.4
PET	4.4	1401	95	9.5	2.16	0.64	87	13.8	1.4
PC	2.4	1201	63	6.3	2.63	0.35	75	9.1	0.9
PMMA	2.1	1179	60	6.0	2.86	0.30	73	8.7	0.9
PP	3.1	900	32	3.2	1.03	0.45	56	4.6	0.5
Acetate	1.8	1320	32	3.2	1.78	0.26	82	4.6	0.5
PE	1.0	930	30	3.0	3.00	0.15	58	4.4	0.4
PVC	2.8	1401	23	2.3	0.82	0.41	87	3.3	0.3

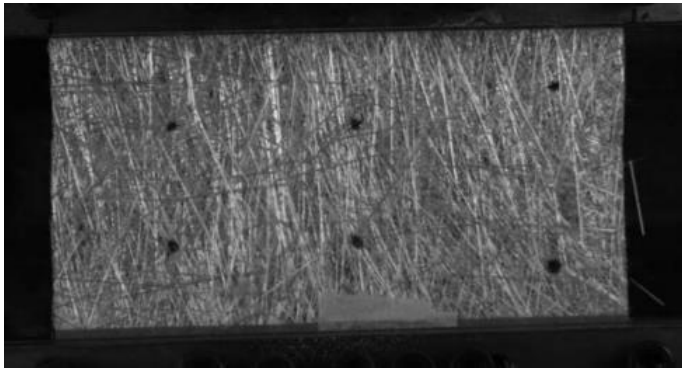
These properties are provided by Oklahoma State University's Web Handling Research Center and included in the WindARoll software included with the book by Keith Good and David Roisum, Winding: Machines, Mechanics, and Measurements.

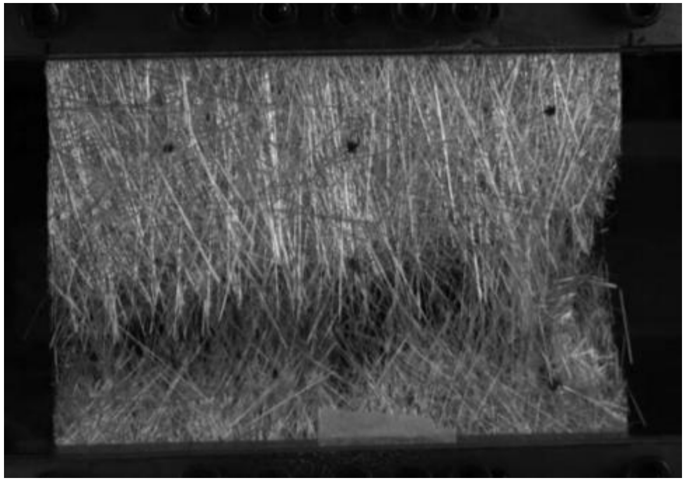
Breakage of Glass Fiber Matting

The mechanical properties of fiberglass matting may not show distinct breakage or yield point. The matting will fail by slippage within the mesh before fiber break. Mechanical behavior will also be dependent on strain rate.



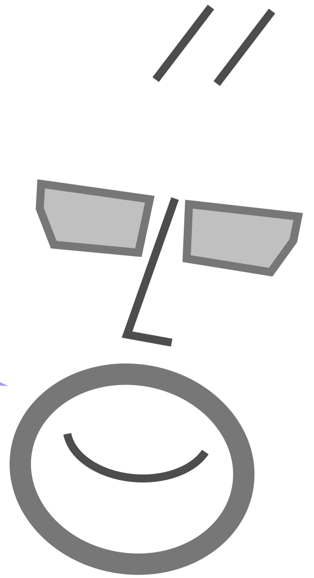






The Right Tension Qs

From data given in the next few slides, what would be the right tension for PET, PP? How about at elevated temperatures?

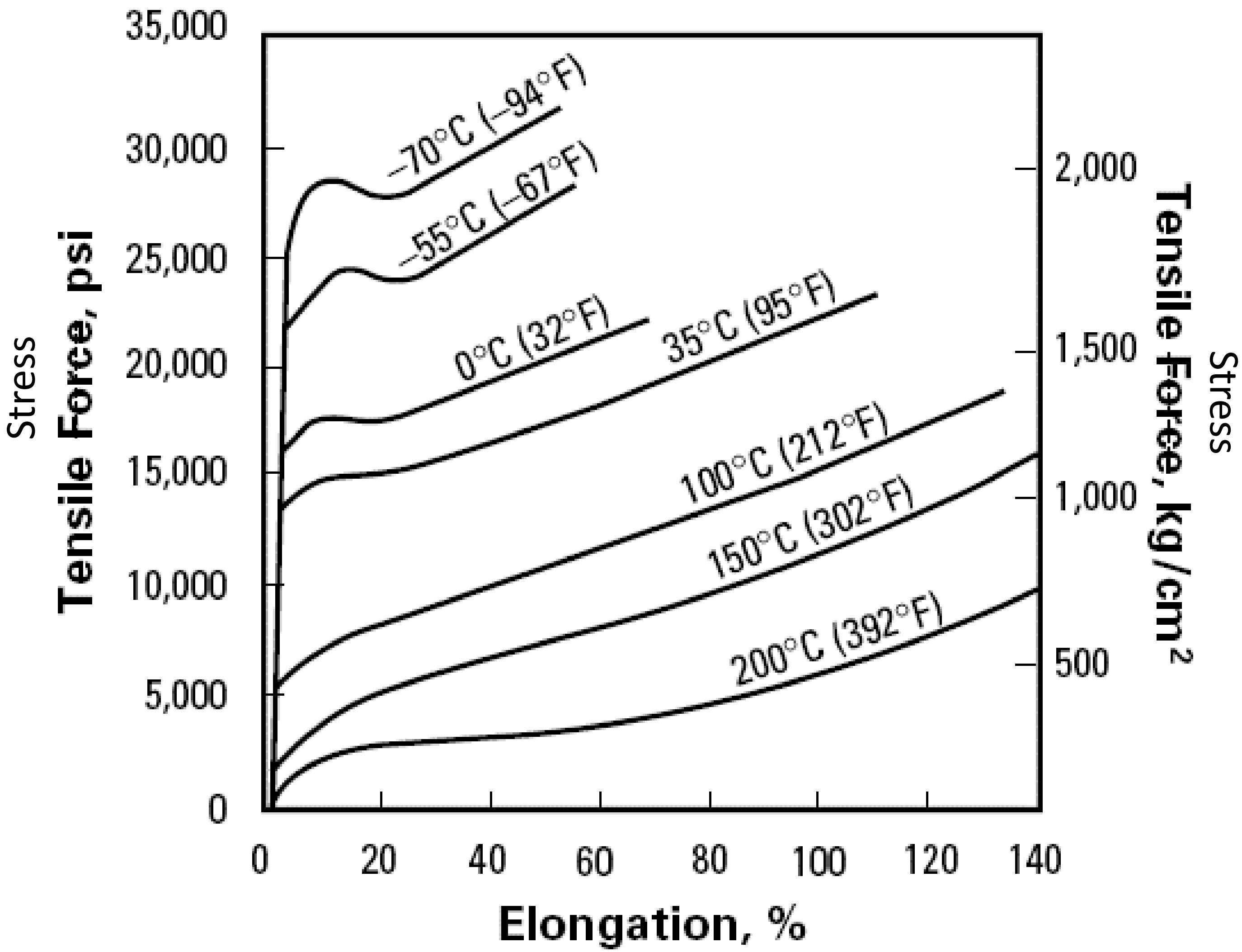


Some Properties of PET

Property	Typical Value	Unit
Gauge and Type **End Use**	**92A** **Industrial**	
Ultimate Tensile Strength, MD TD	20 (29) 24 (34)	kg/mm² (kpsi)
Strength at 5% Elongation (F-5), MD TD	10 (15) 10 (14)	kg/mm² (kpsi)
Modulus, MD TD	490 (710) 510 (740)	kg/mm² (kpsi)
Elongation, MD TD	116 91	%
Surface Roughness Ra	38	nm
Density	1.390	g/cm³
Coefficients of Thermal Expansion	1.7×10^{-5}	in/in/°C
Melt Point	254	°C
Dimensional Stability at 105°C (221°F), MD TD at 150°C (302°F), MD TD	0.6 0.9 1.8 1.1	%

From DuPont Teijin Films Mylar® Data Sheet

Yield Stress and Temperature

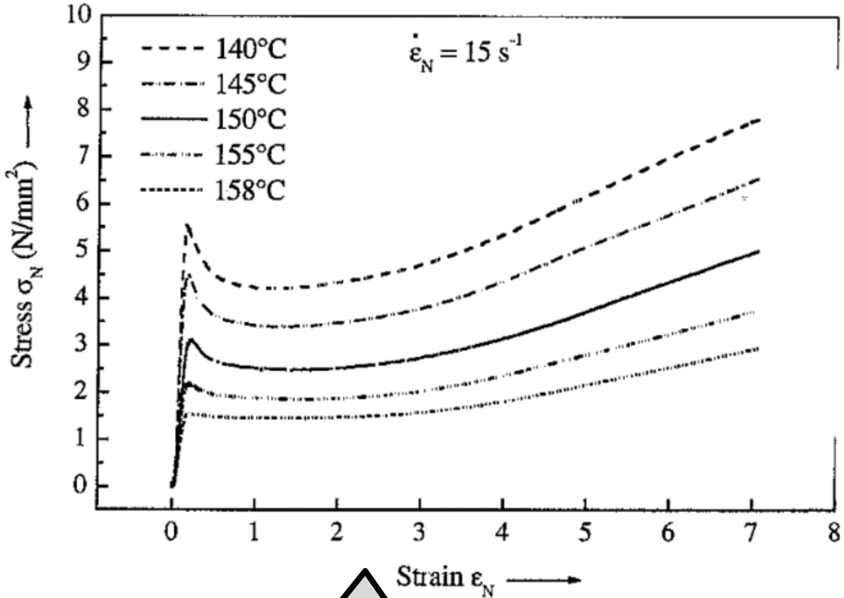


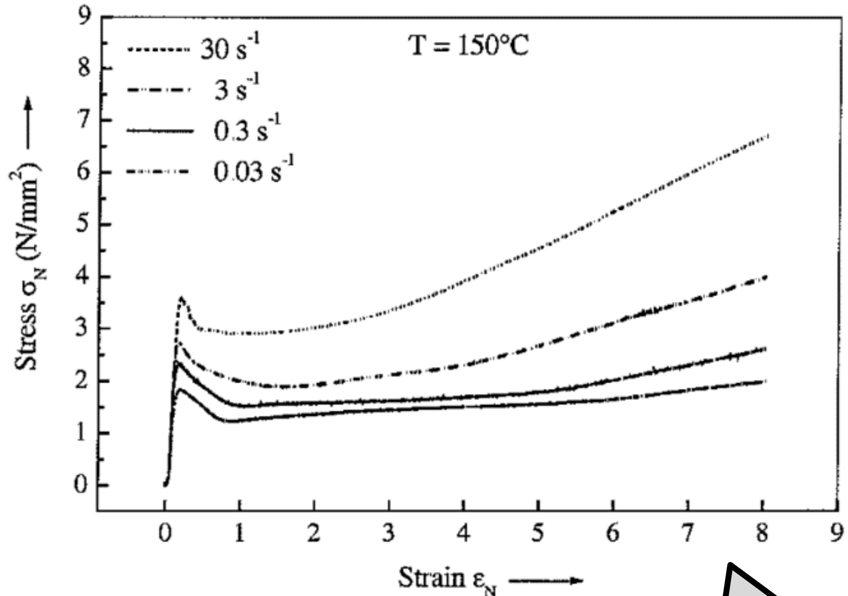
From DuPont Teijin Films Mylar® Data Sheet

Polypropylene Mechanical Properties

PP-Q: Water quenched
PP-S: Slowly quenched
PP-N: Nucleating additive

Cast Film	Nucleating Agent	Extrusion Temp. (°C)	Chill-Roll Temp. (°C)
PP-Q	-	250	20
PP-SC	-	250	80
PP-N	2 wt.%	250	20





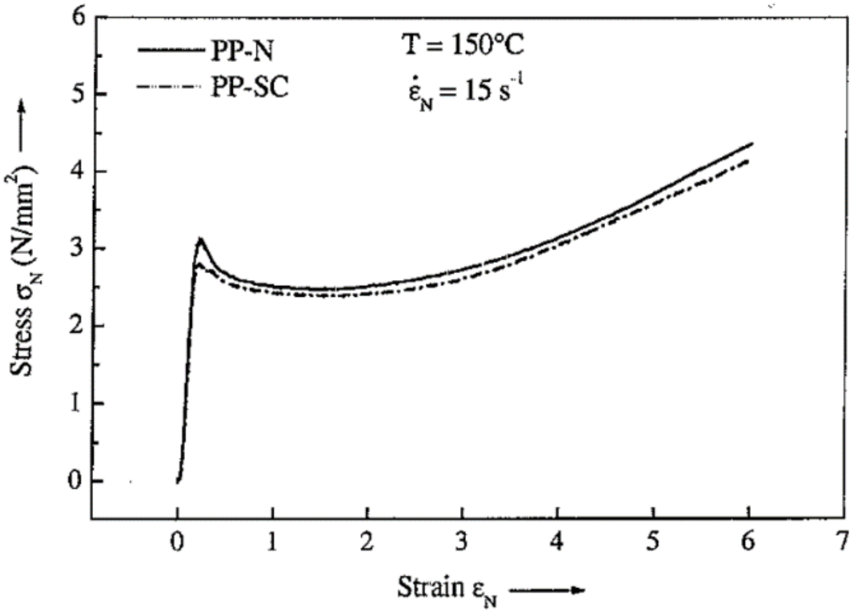
Effect of Temperature

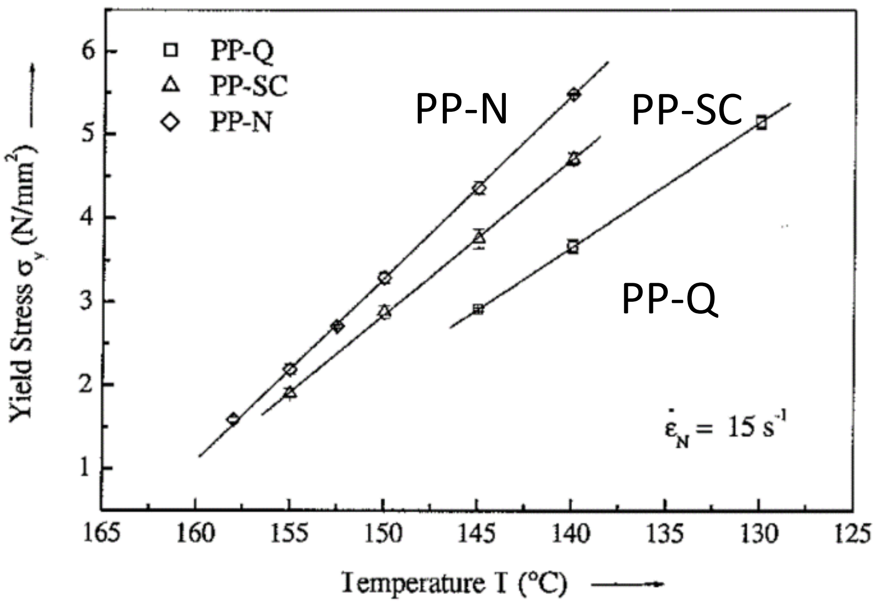
Effect of Strain Rate

Deformation Behavior of Different Polypropylene Cast Films at Elevated Temperatures

S. Rettenberger (1), L. Capt (1), H. Münstedt (1), K. Stopperka (2), J. Sänze (2)
(1) University Erlangen-Nuremberg, Institute of Polymer Materials, Department of Materials Science
Martensstrasse 7, 91058 Erlangen, Germany
(2) Brückner Maschinenbau GmbH,
P.O. box 1161, 83309 Siegsdorf, Germany

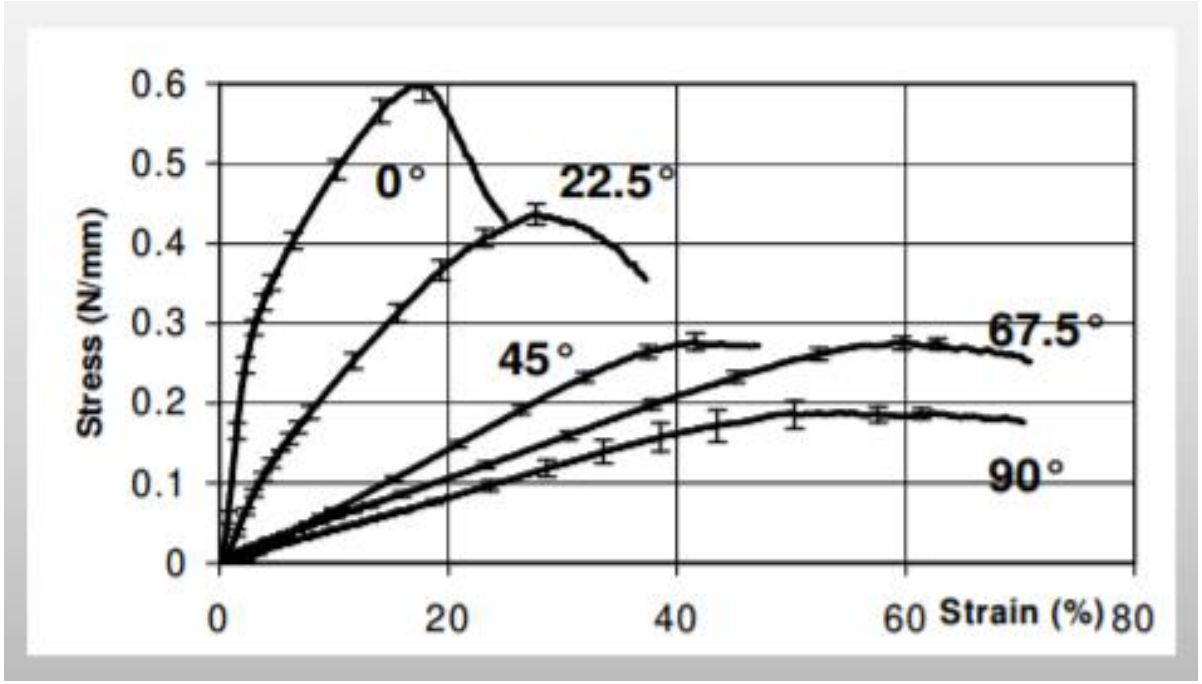
Differences in Polypropylene by Type and Temperature





Mechanical Properties of Nonwoven

Stress-strain behavior of through air bonded nonwoven. Like polymer films, the effect 'modulus' drops with higher temperatures.



Mechanical Characterisation of Through-air Bonded Nonwoven Structures

Amit Rawal, Stepan Lomov, Thanh Ngo, Ignaas Verpoest and Jozef Vankerrebrouck

Moisture and Temperature

Webs change thickness, width, and length with changes in moisture content and temperature.

Dimensional changes of a magnitude similar or larger than web strain will significantly change roll structure.

Courtesy of Machine Design magazine

Buckling defects from hygroscopic expansion of paper rolls.

Courtesy of RTape

Wind, store, and unwind roll at stable humidity and temperature conditions, optimally 40-60% relative humidity and 65-75°F.

Hygro-Expansion of Papers

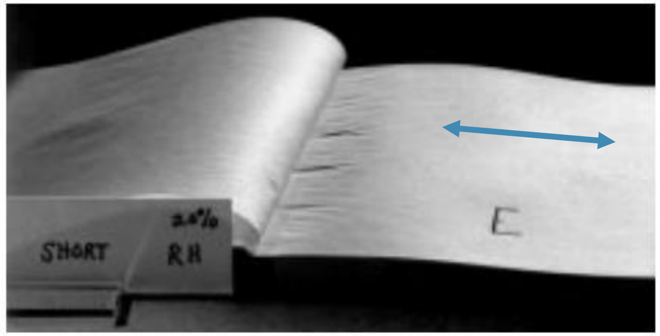
Paper in all four books was 20% RH at the time of binding.

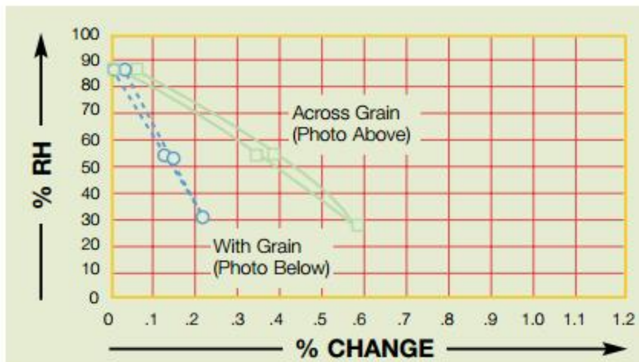
Books at top were bound perpendicular to paper grain (across grain edge)

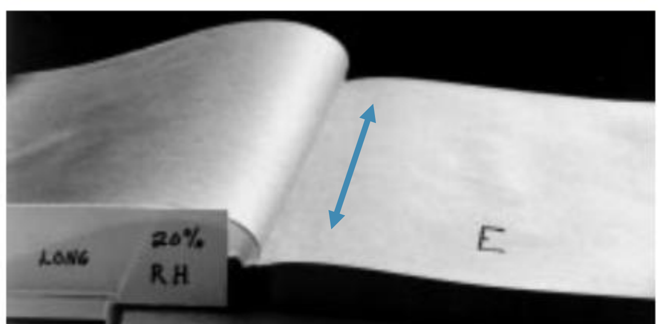
Books on the bottom were bound parallel to paper grain direction.

Graphs in center are “hygroexpansivity” graphs showing percentage of size change at different RH equilibrium points for each paper.

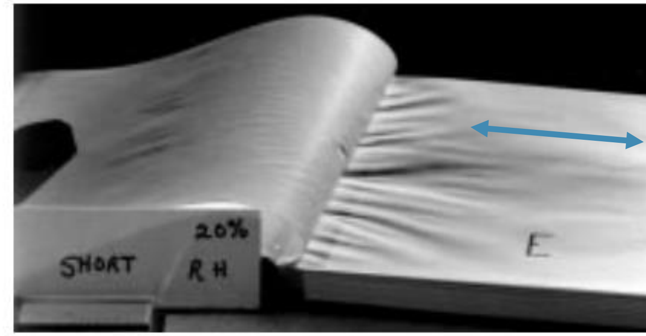
Newsprint

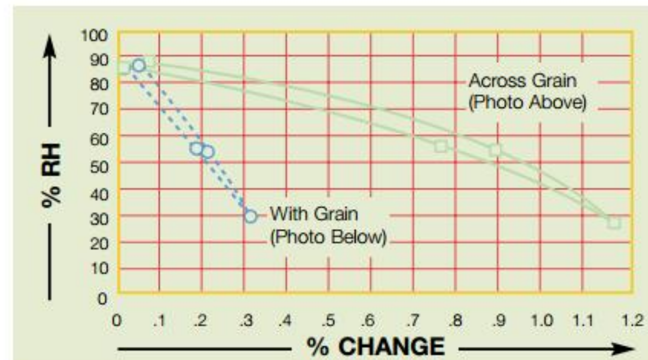


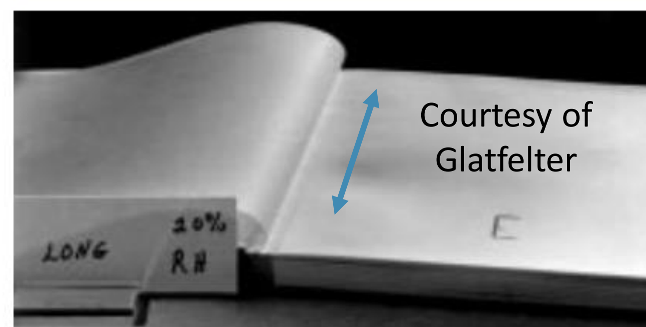




Coated, Not Calendered







Courtesy of Glatfelter

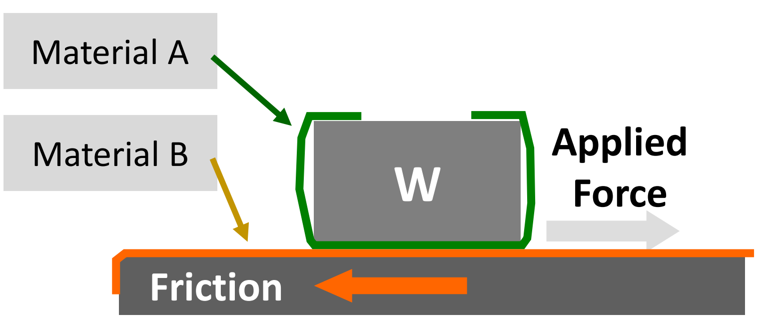
Coefficient of Friction (COF)

The coefficient of friction, $\mu_{A\text{-}B}$, is the ratio of frictional force relative to normal load between surface A and surface B. COF is a property of the combination of two material surfaces.

Two common friction coefficient measurement methods are:

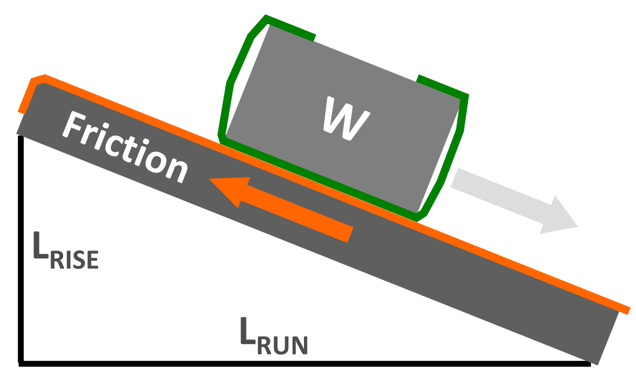
Sliding Block Test

$$\mu_{A-B} = \frac{F}{W}$$



Incline Plane Slide Test

$$\mu_{A-B} = \frac{L_{RISE}}{L_{RUN}}$$



COF Tester


TMI
Slip and Friction Tester
Model 32-07
TESTING MACHINES INC.
www.testingmachines.com

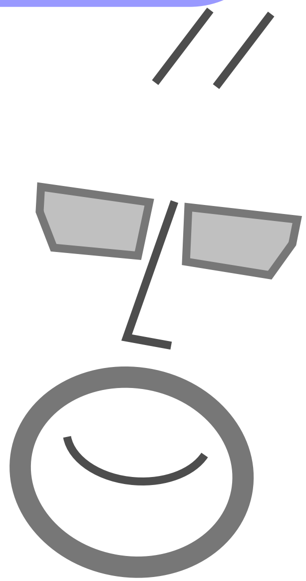

TMI Model 32-25
COEFFICIENT OF FRICTION TESTER

The Right Tension Qs

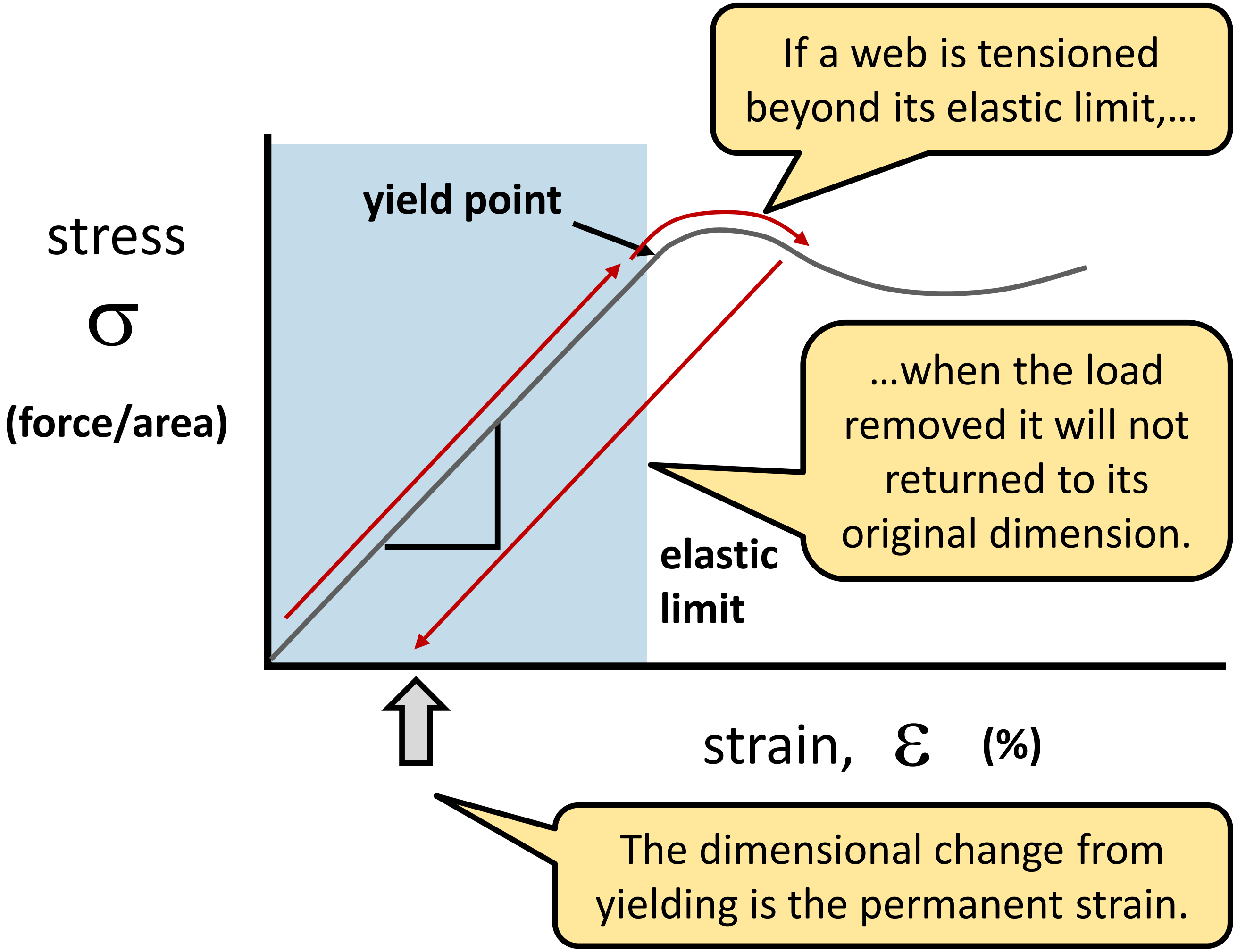
What is elasticity?

What is the elastic limit?

What is visco-elasticity?



Tensioning Beyond the Elastic Limit



Webs can yield in all three dimensions, creating permanent changes in length, width, or thickness.

KEY CONCEPT

Elastic vs. Viscoelastic Behavior

Elastic materials respond to stress with an immediate strain change and recover to initial dimensions when stress is removed. Viscoelastic materials will have a time-dependent response to stress or strain.

Use: Most webs can be considered elastic.
Example: Tensioning a web will create an immediate stretch (a.k.a. strain), that is independent of time or speed.

Use: Under constant load, a viscoelastic webs will continue to stretch over time.
Example: Hang a weight on a strip of vinyl electrical tape and measure the length over time.

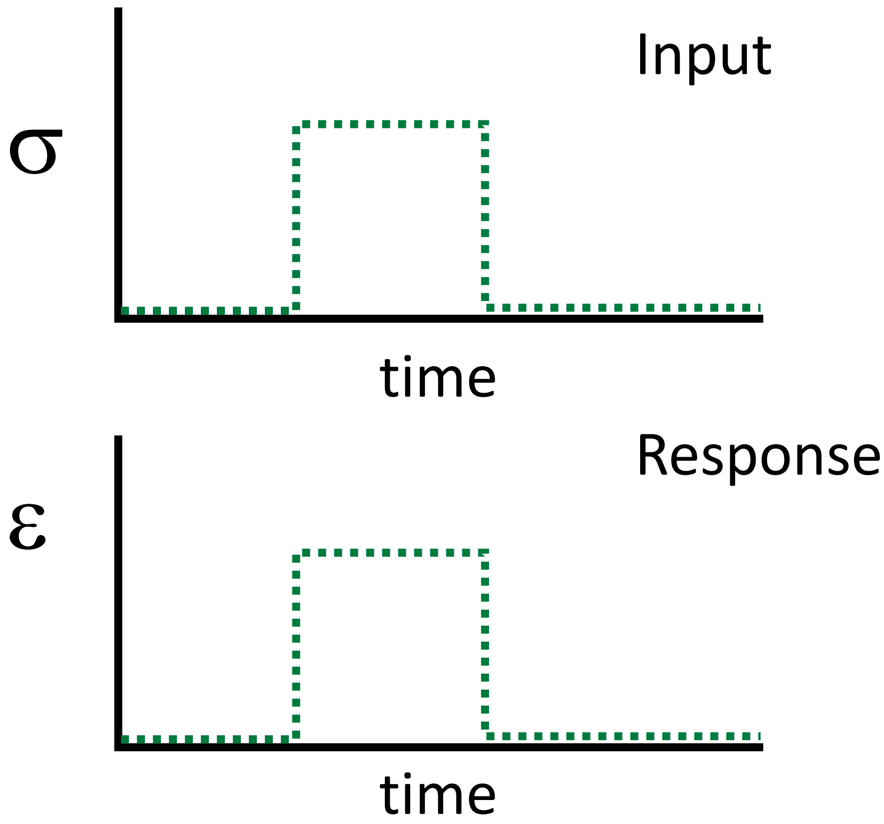
Elastic Behavior

Elasticity

Elastic materials will respond proportionally and immediately (at the speed of sound).

Applied stress will result in web strain.

Applied strain will result in web stress.



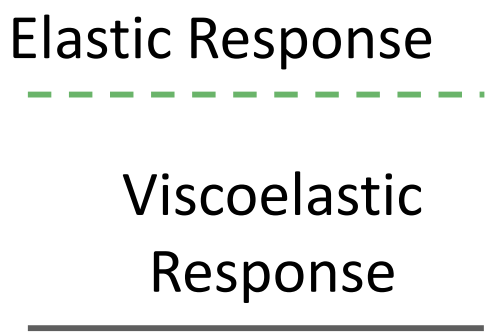
Elastic Response

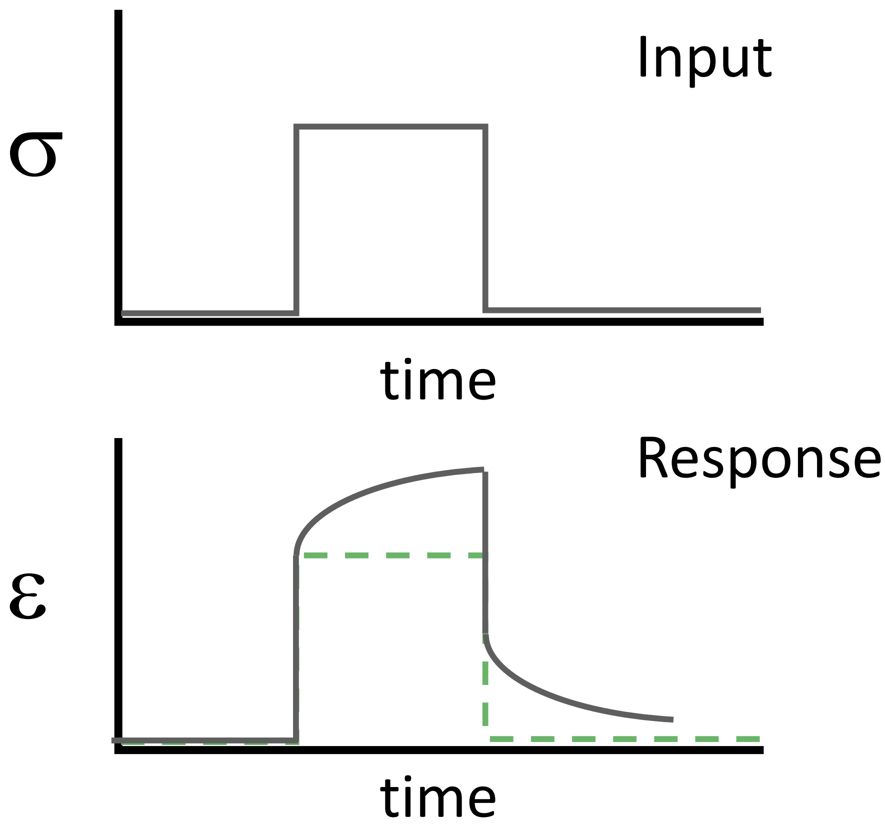
Viscoelastic Behavior - Creep

Creep Test

A constant stress is applied to the web.

Response is strain.





VE Behavior – Stress Relaxation

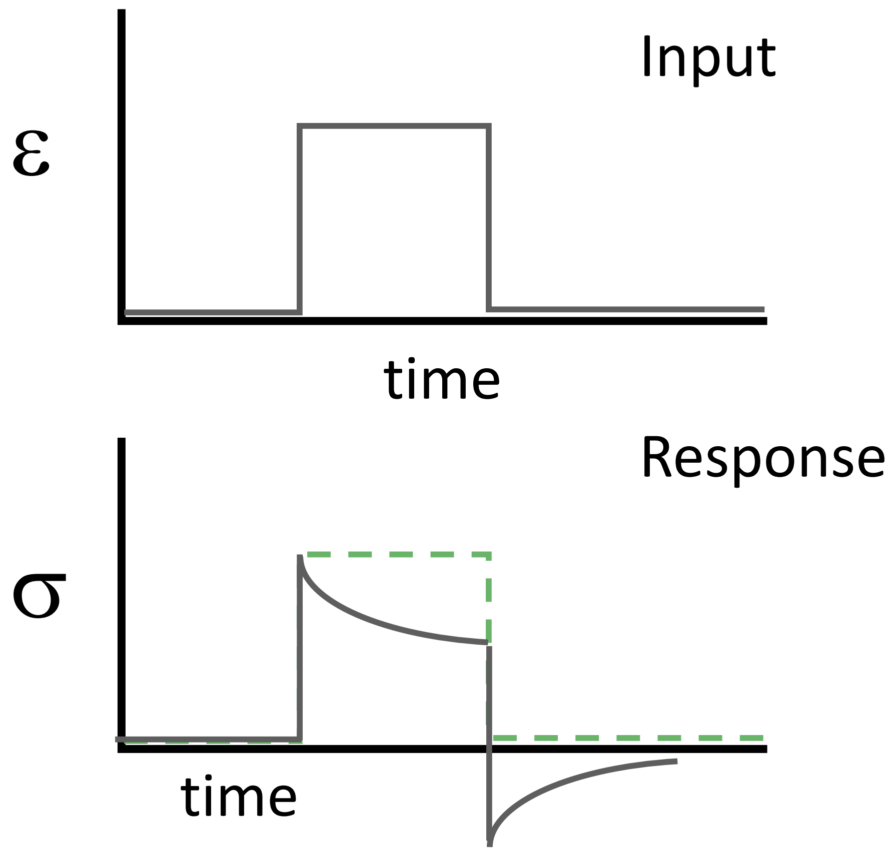
Stress Relaxation

A constant strain is applied to the web.

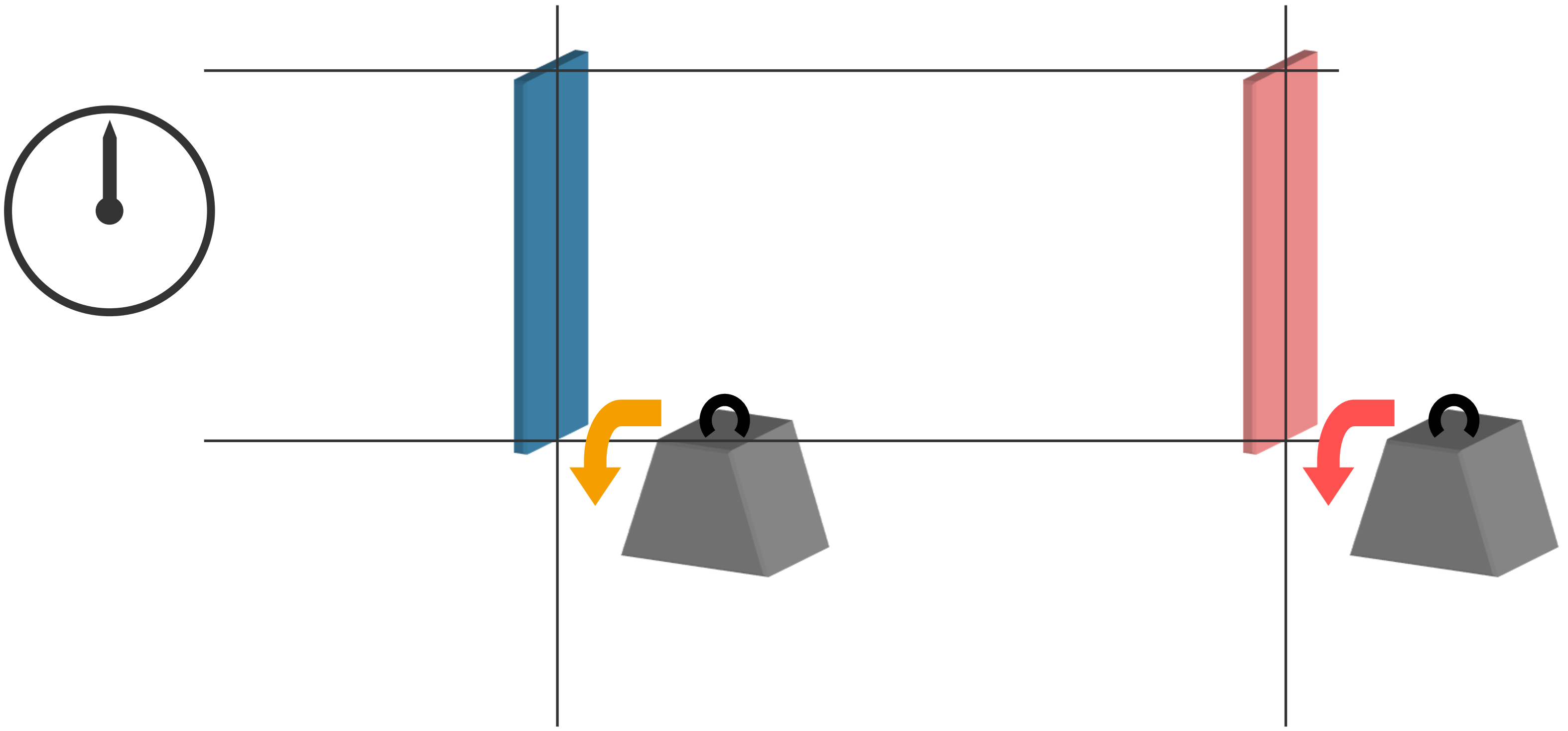
Response is stress.

Elastic Response

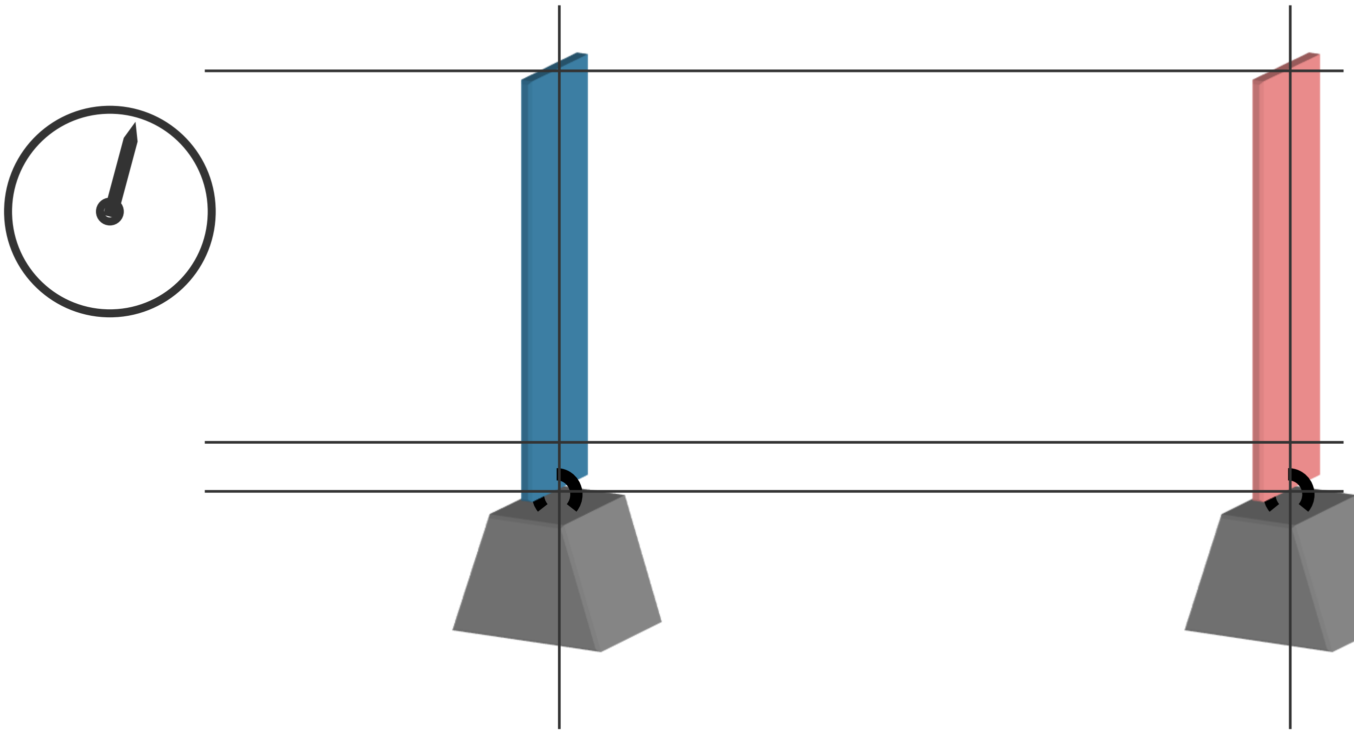
Viscoelastic Response



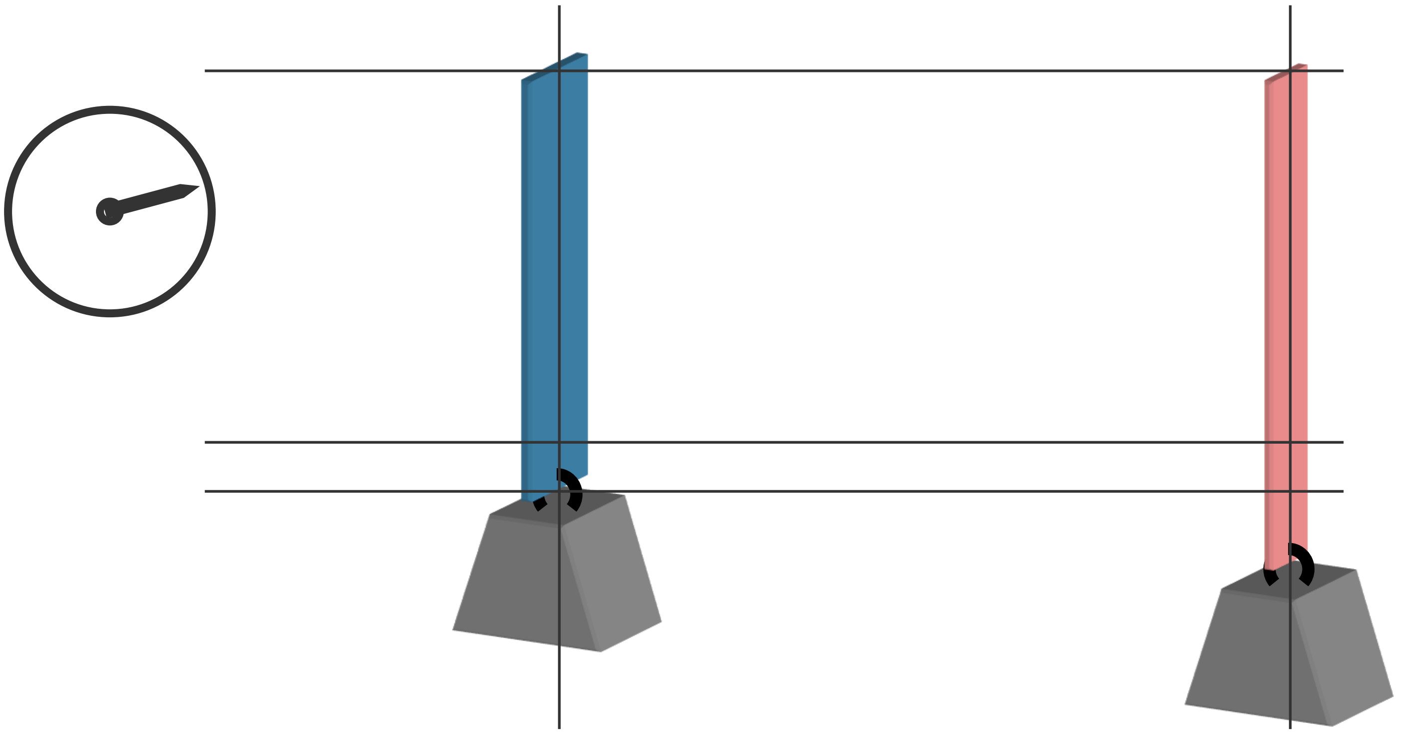
Viscoelastic Behavior - Creep



Viscoelastic Behavior - Creep



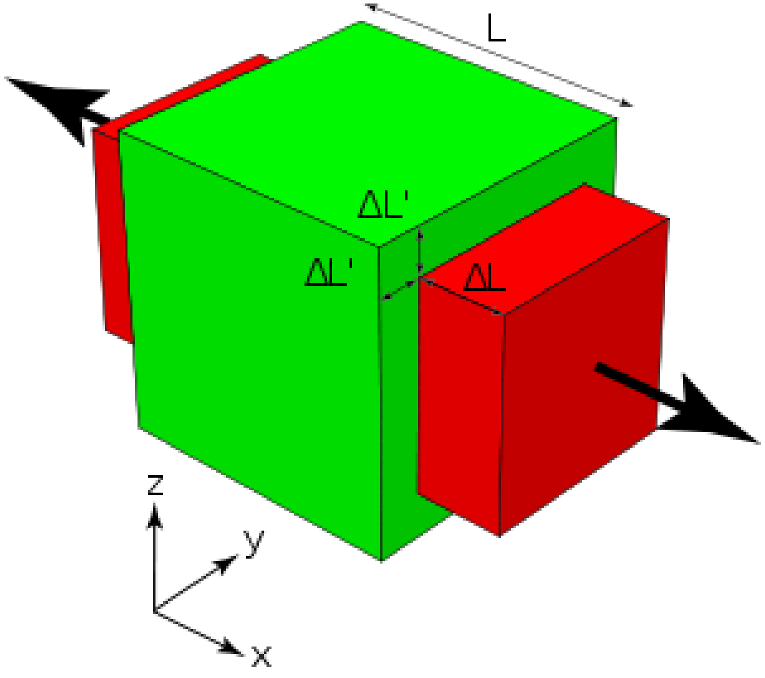
Viscoelastic Behavior - Creep

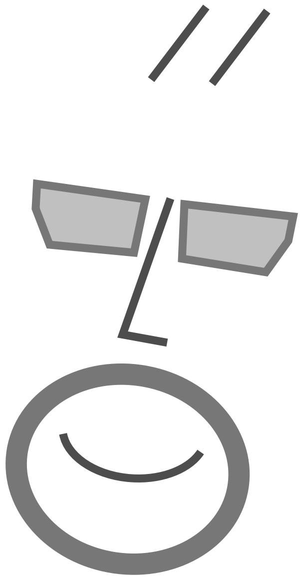


The Right Tension Qs

TJWALKER+ ASSOCIATES INC

What is the Poisson's ratio?

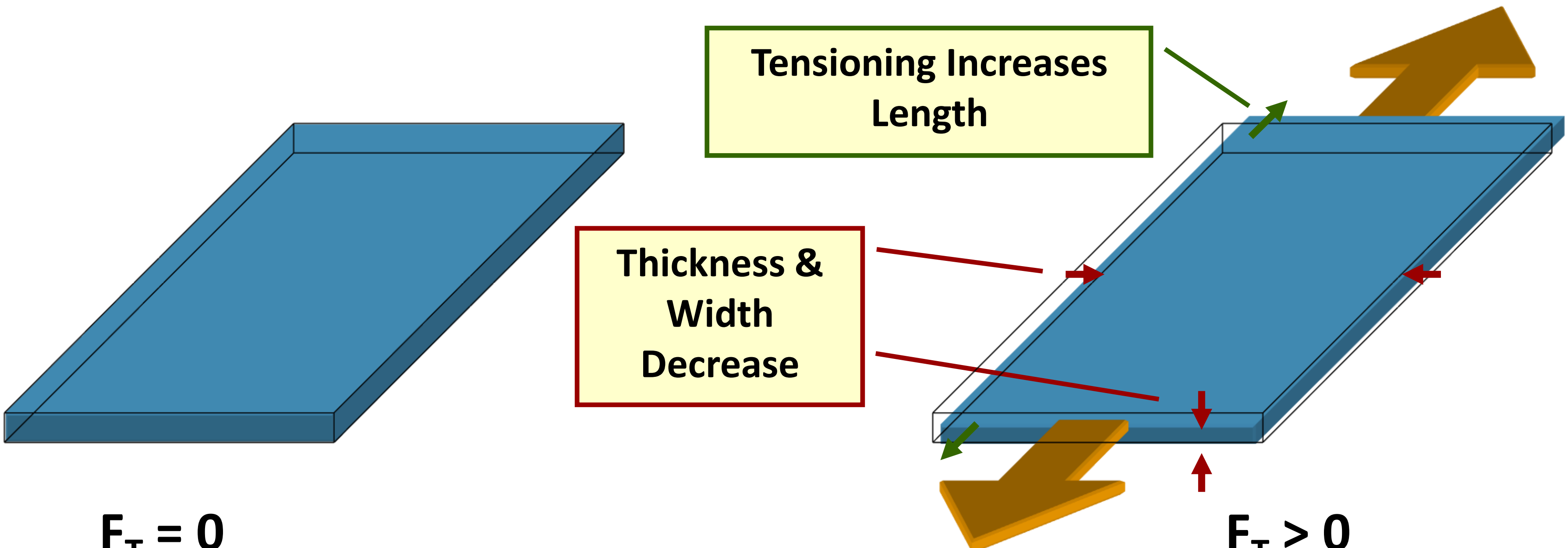




Strain = Dimensional Change

For solid materials, tensile and compressive stresses do not significantly change density.

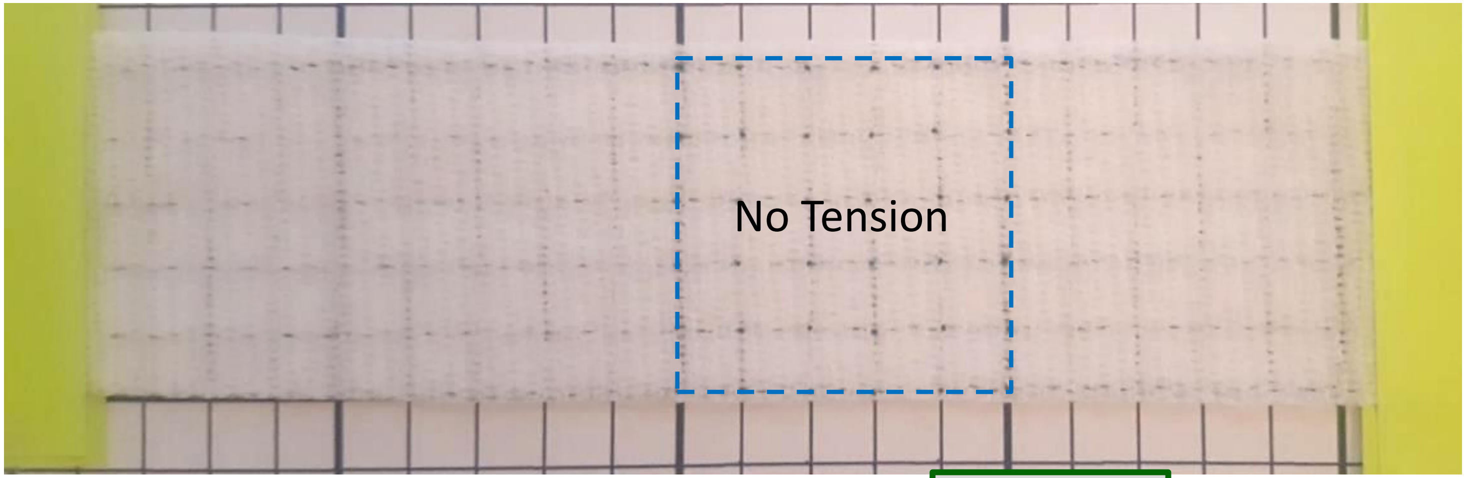
Increases in length (MD strain) is offset by decreases in the width and thickness.



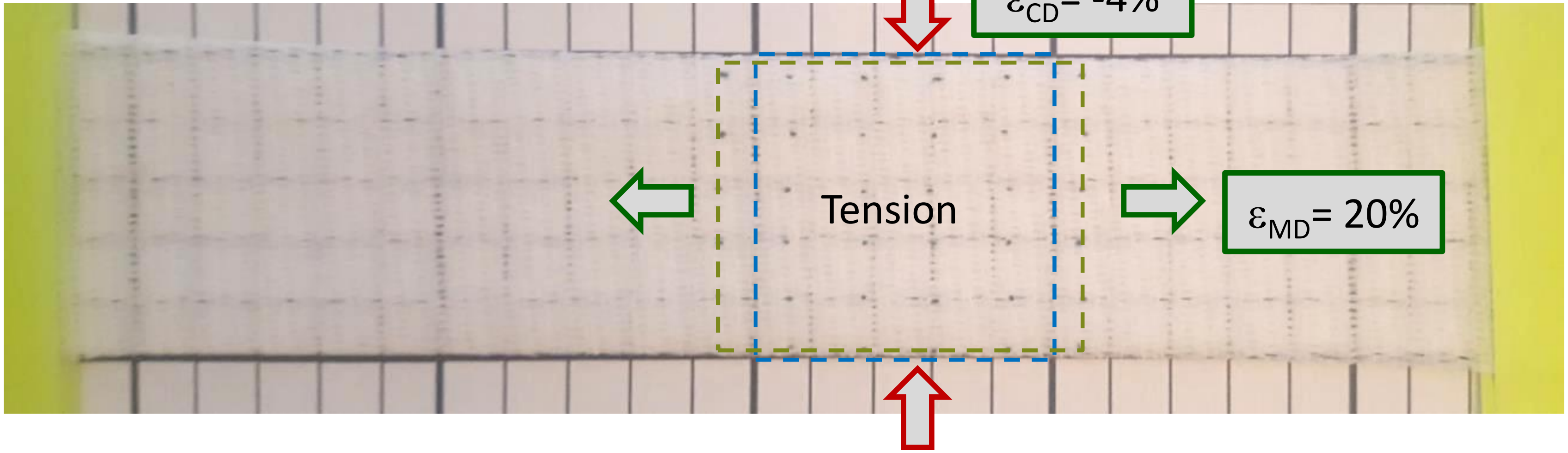
$F_T = 0$

$F_T > 0$

Example of Poisson's Ratio



ε_{CD}= -4%



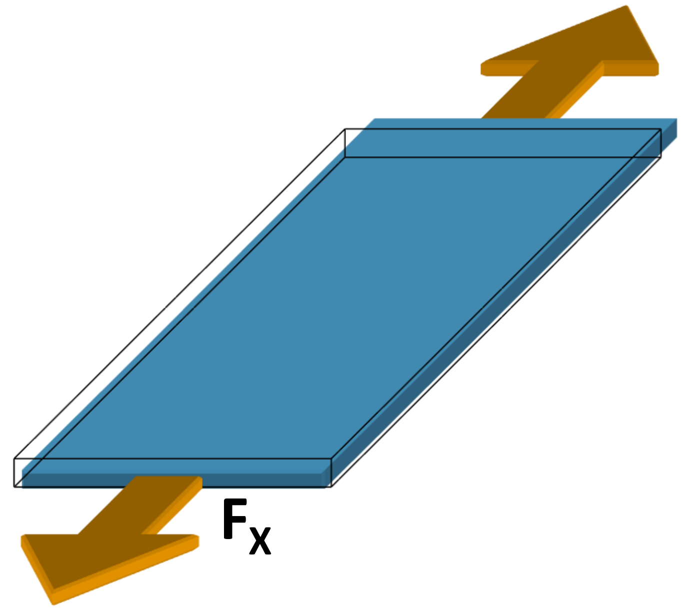
ε_{MD}= 20%

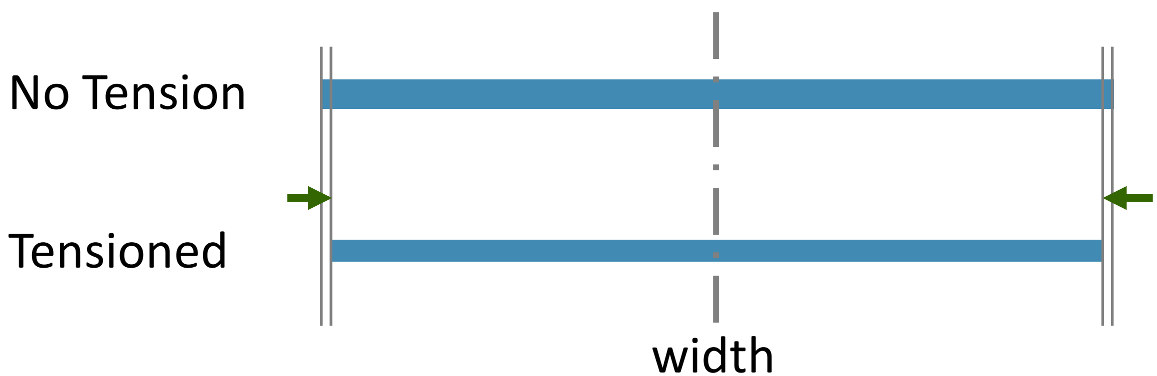
CD Contraction

Tension, thickness, width, modulus (stretchiness) and Poisson's ratio determine how much the web will contract.

$$\varepsilon_Y = -\nu_{XY}\varepsilon_X = -\nu_{XY}\left(\frac{F_X}{twE}\right)$$

$$\delta w_T = w_0\left(-\nu_{XY}\varepsilon_X\right)$$

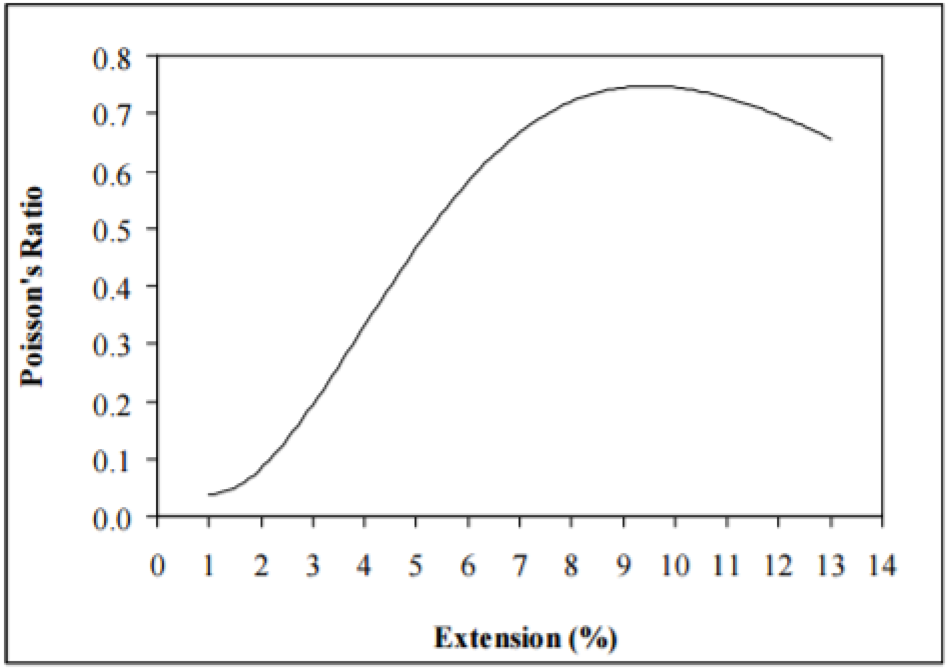




Poisson's Ratio of Fabrics

The Poisson's ratio of a fabric is not a constant. For initial strain (extension), the Poisson's ratio is near zero.

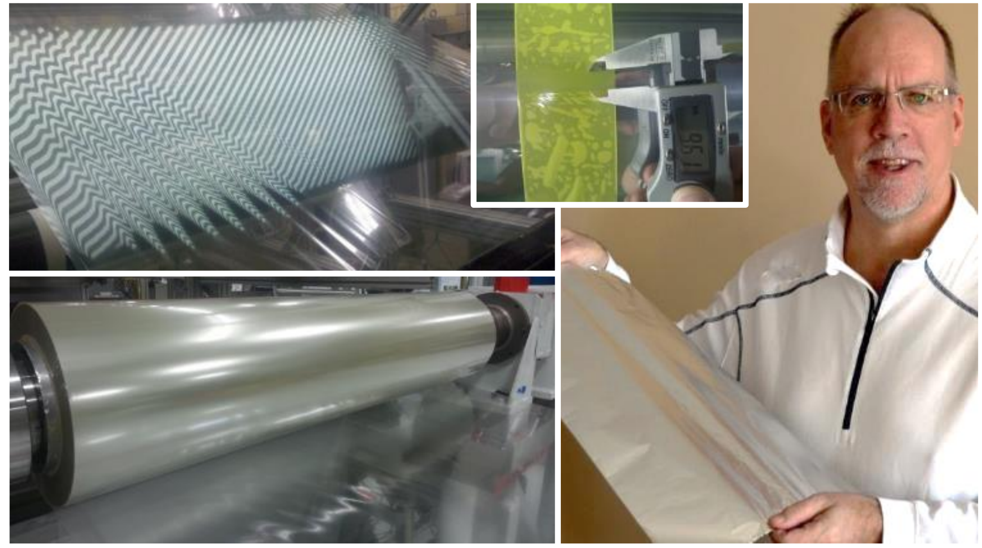
As strain increases, the Poisson's effect increase up to a peak value.



Effect of Fabric Structure and Weft Density on the Poisson's Ratio of Worsted Fabric

Nazanin Ezaz Shahabi, Siamak Saharkhiz, PhD, S. Mohammad Hosseini Varkiyani, PhD

Textile Engineering Department, Amirkabir University of Technology, Tehran IRAN



1.2 Bagginess and Curl

What is…? What causes…?

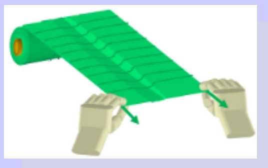
Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

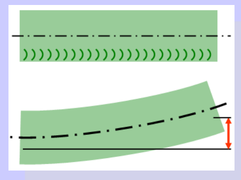
tjwalker@tjwa.com www.webhandling.com 651.686.5400

Solutions to Imperfect Webs

Cambered and Baggy Webs



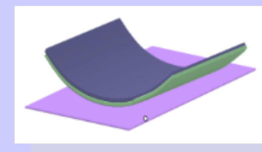
Baggy Webs



Cambered Webs

Bagginess and camber are often strongly related to winding.

Non-Flat Webs



Curled Webs

Curl is strongly related to coating and laminating (and sometimes winding).

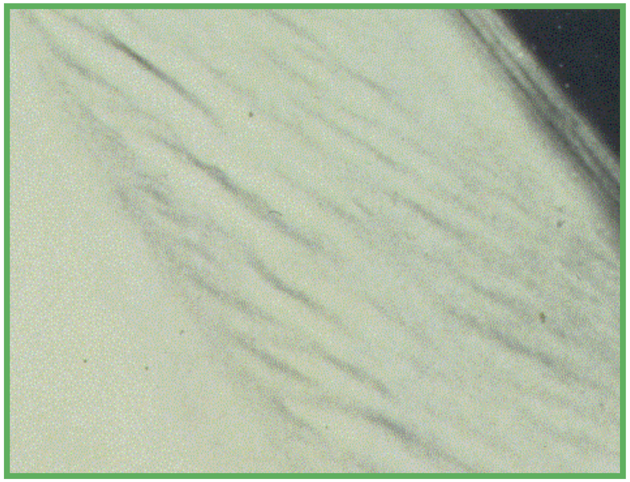
Imperfect Web Solutions Questions

What is web bagginess?
What is web camber?
What is web curl?

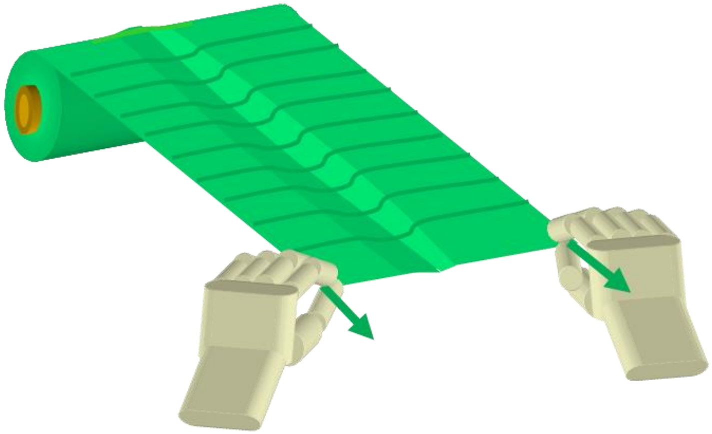
How are they measured?

2A: Baggy Web

Definition:
A web with crossweb length variations. A web is considered baggy if the strain of tension does not pull the web to flatness.



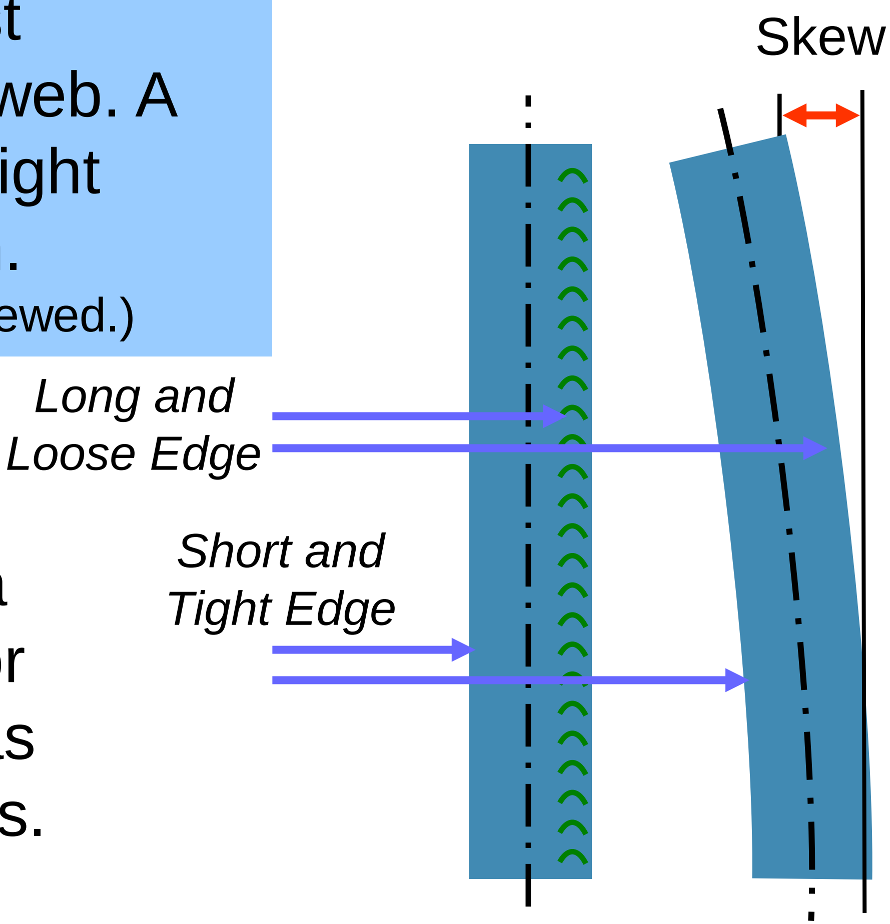
Measuring Bagginess:
- There are many methods to qualify or quantify bagginess.
- Most are difficult, expensive, or time-consuming.
- A manual pull out and 1-to-5 grading may be the simplest.
- For films, roll hardness can be a useful predictor or bagginess.



2B: Skew/Camber

Definition:
Skew (a.k.a. Camber) is the most common term for a non-straight web. A cambered web will bend left or right when rolled out under no tension.
(Slit strands cut from baggy web may be skewed.)

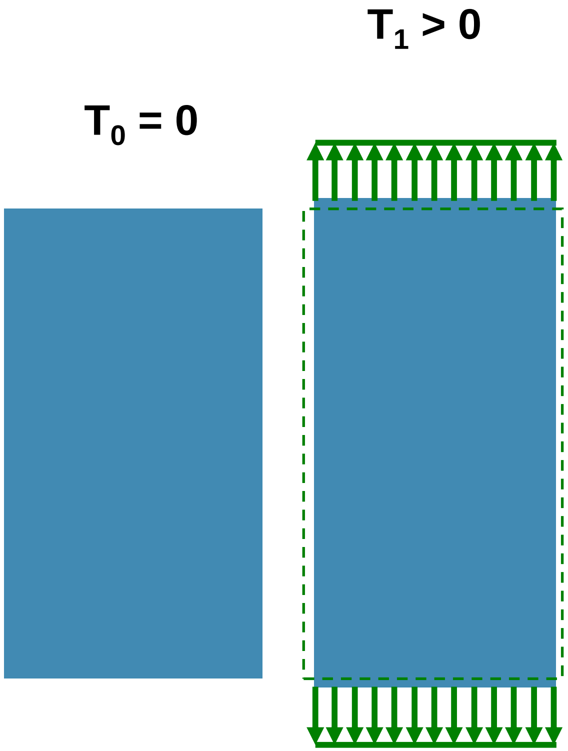
Measuring Skew:
Skew is typically measured by pulling, rolling, or sweeping out a long sample on a tabletop or floor and quantifies the left or right bias of the web relative to straightness.

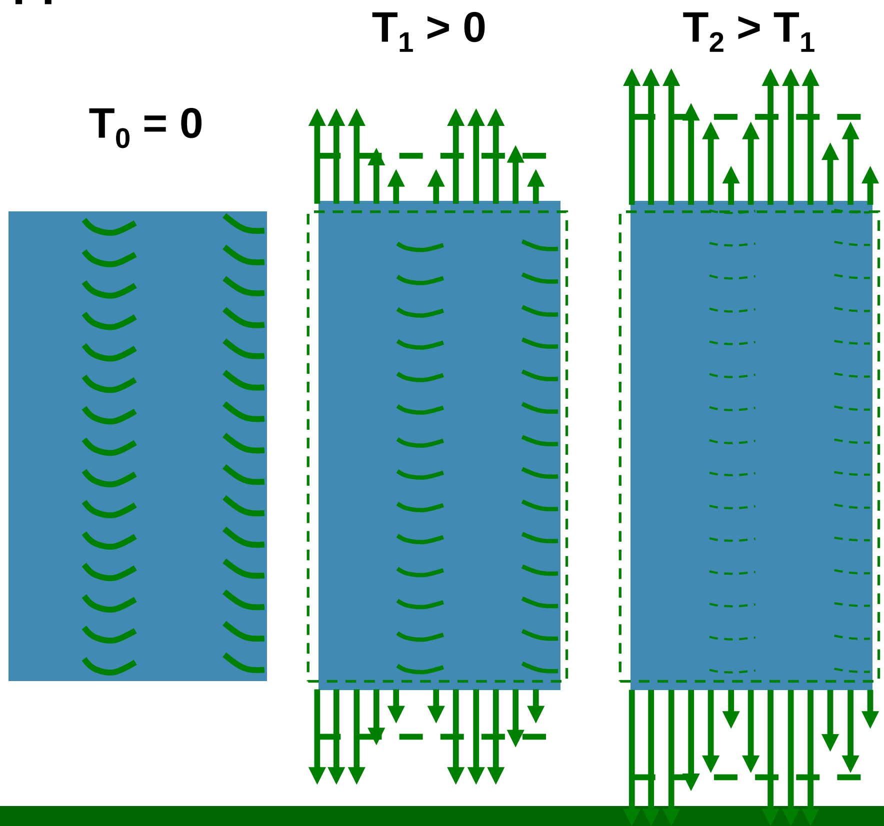


Tensioning Baggy Webs

The ideal web carries tension uniformly across the web width.

For an imperfect web, tension stretches the short lanes first. When short lanes are stretched to equal the long lanes, the web appears taut.

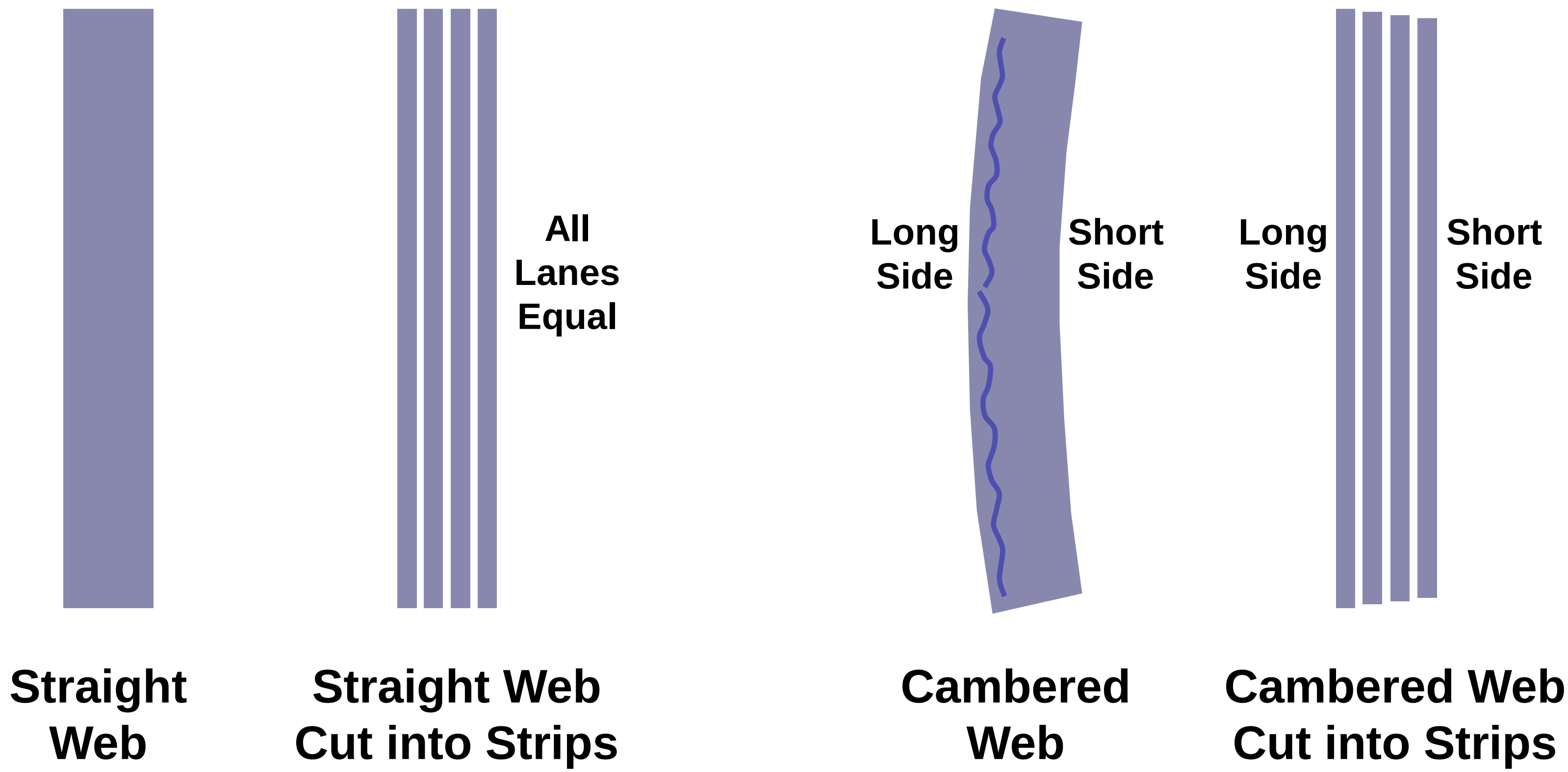




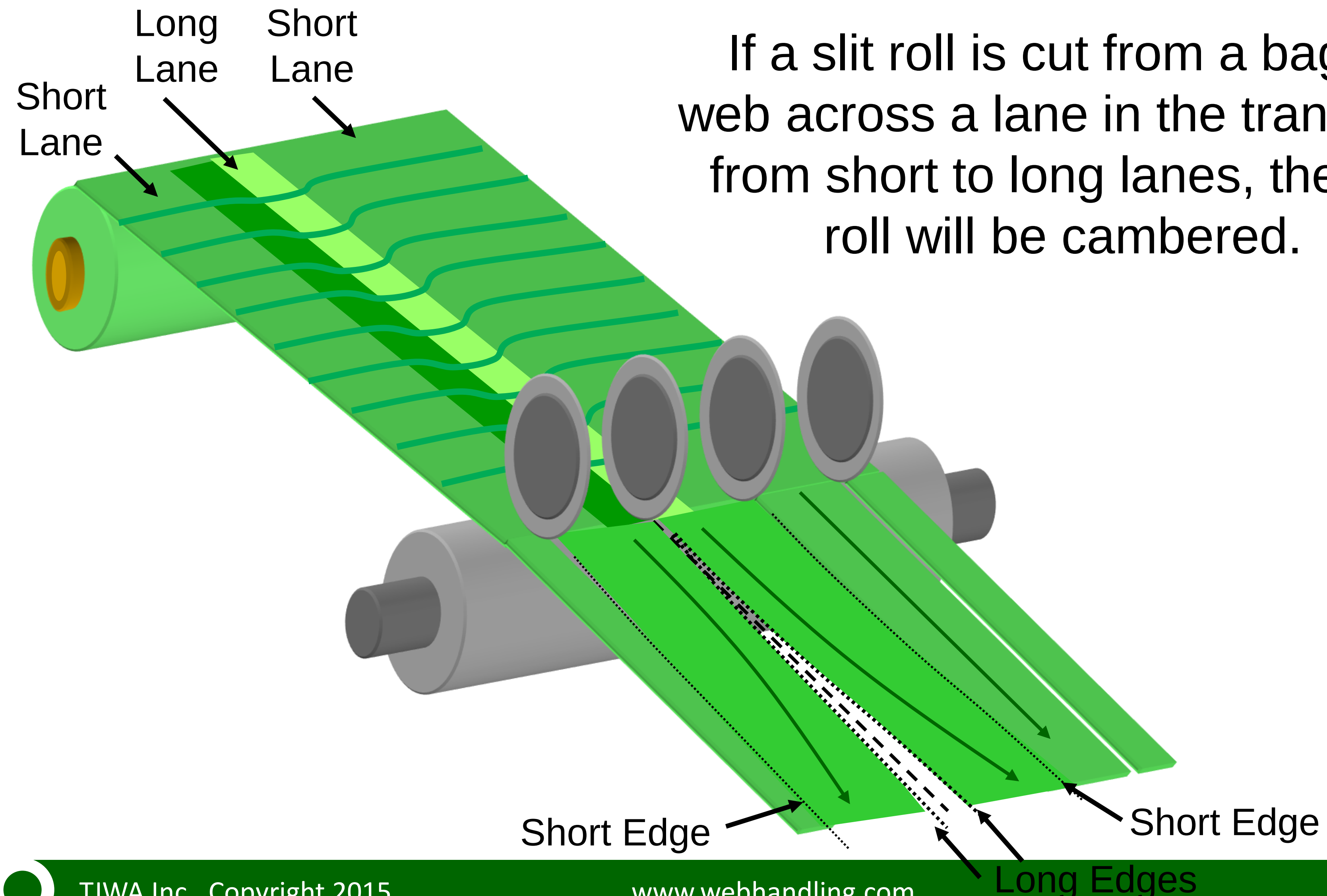
Cambered Web (a.k.a. Skewed Web)

A cambered or skewed web has one side longer than the other.

A cambered web will curve left or right when rolled or swept out on the floor.



Slitting Baggy or Cambered Web



If a slit roll is cut from a baggy web across a lane in the transition from short to long lanes, the slit roll will be cambered.

Imperfect Web Solutions

What causes CD length variations in paper, film, or foil manufacturing ?

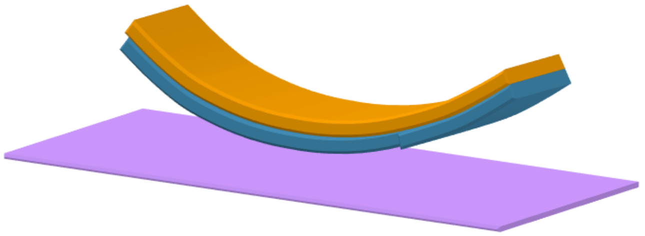
A short, but not complete list why paper, film, and foil making has CD length variations includes:

- *Machine misalignment*
- *Crossweb moisture variations*
- *Uneven calendaring*
- *Variation in roller diameters*
- *Uneven web paths or nip loads in blown film collapsing*
- *Cooling or stress variations in tenter quenching zone*
- *Nip variations in laminating or film extrusion nips*

2C: Web Curl

Definition:
Curl is the tendency for a web to not lay flat (especially coated or laminated webs), instead form into a curved or scroll shape.

Measuring Curl:
Curl is best measured in narrow strip samples. Strips samples are usually cut in the machine or cross machine direction, but may be cut at any angle.



More on Curl... More on Bagginess...

More on curl in Section 4
Nipped Rollers and Laminating

More on bagginess in seminar
on Winding and Roll Quality (6)